Spillover Effects of Financial Incentives for COVID-19 Vaccination: A Population-level Cluster Randomized Controlled Trial

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Key Points

Question: Are there negative consequences of using financial incentives to encourage vaccination?

Findings: In a population-level randomized controlled trial (RCT), financial incentives had no direct impact on the recipients' likelihood to get vaccinated against COVID-19. Individuals living in the same household as financial incentive recipients were 22.8% less likely to receive a COVID-19 booster as compared to their counterparts in the control group; this -0.3 percentage point spillover effect (n_{treat}=154, $n_{control}$=199) was statistically significant.

Meaning: Our results illustrate the importance of incorporating measures for spillover in RCT designs to avoid costly, ineffective, or even counterproductive policy outcomes.

Abstract

Importance: Prior studies show that financial incentives have small positive direct effects in increasing COVID-19 vaccinations, however estimates of social and temporal spillovers are almost entirely absent from the existing literature.

Objective: Building on a previously-successful COVID-19 financial incentive, we sought to assess whether incorporating estimates of social and temporal spillover in a research design meaningfully affects big-picture evaluations of policy effectiveness.

Design: In a population-level, address-cluster RCT, participants were randomized to treatment and control, with one resident in each address-cluster randomly selected to be an address-cluster representative. The address-cluster representatives in treatment addresses received the treatment letter; all others received the control letter.

Setting: Late 2021, Ravensburg, Germany.

Participants: All adult residents of Ravensburg, Germany: 10,032 address-clusters or 41,548 residents in total.

Intervention: The control letters informed recipients about seven upcoming free COVID-19 vaccination events. The treatment letters were identical to the control letters, except they also offered recipients 40 Euros should they get vaccinated at one of the vaccination events.

Main Outcome(s) and Measure(s): Primary and booster vaccinations were recorded during the events.

Results: We estimated the direct effect of the treatment letter on the recipient, the social spillover effect (i.e., impact on the cohabitants of treatment letter recipients), and the overall (address-level) effect. The direct, spillover, and overall effects on primary vaccinations were all indistinguishable from zero (0.1pp, 95% CI [-0.22, 0.33], p=.69; 0.0pp, [-0.22, 0.26], p=.86; 0.0pp, [-0.17, 0.23], p=.76, respectively). For boosters, the direct and overall effects were *negative* (-0.3pp, [-0.77, 0.14], p=.17; -0.3pp, [-0.51, -0.09], p<.01, respectively), and the spillover effect was *negative* and statistically significant (-0.3pp, [-0.53, -0.06], p=.01).

Conclusions and Relevance: Financial incentives for vaccination can have negative impacts on those who don't receive the incentives. Based on the timing of the effect, there is suggestive evidence that individuals who resided with a recipient of the financial incentive chose to postpone booster vaccination.

Trial Registration: DOI:10.1186/ISRCTN59503725

Introduction

The importance of behavioral science in public health was made particularly prominent during the COVID-19 pandemic. While public health and medical researchers saw vaccination as the best means of responding to COVID-19 [1,2], many policy-makers around the world struggled to convince people to get vaccinated [3–5]. While the COVID-19 pandemic was a particularly high profile example, it was reflective of broader trends in public health and policy contexts: policy-makers increasingly leverage behavioral science to bridge medical research and policy.[6,7] However, when behavioral science interventions are scaled up, temporal and social spillovers could plausibly have serious unforeseen consequences [8]. By social spillovers, we mean any changes in outcomes among individuals who are not direct beneficiaries of a policy but are socially exposed to policy beneficiaries. By temporal spillover, we mean any policy impacts that occur after initial outcomes are collected. If these spillovers are left unmeasured, policy-makers may erroneously believe their programs are effective and evidence-based when, on the whole, they are ineffective or, in some cases, even harmful. Such policy spillovers may be extremely challenging to anticipate prior to policy implementation (e.g., Bouckaert et al.[9]).

We illustrate the risk of leaving spillovers unmeasured via a novel, population-level RCT that used a popular[10,11] intervention to encourage vaccination against COVID-19. One of the most prominent field RCTs at the time we were designing our study found that, among a sample of online survey respondents, offering monetary incentives of about $24 to survey-takers increased vaccination uptake by about 4 percentage points[12]. The same research team later addressed the possibility that financial incentives could lead to a backlash in the future and concluded that "financial incentives for vaccination do not have negative unintended consequences"[13(p1)]. However, as is common practice in policy RCTs, neither the original nor the expanded study looked at social spillover. Of the existing literature on monetary incentive effectiveness for COVID-19 vaccination, a recent meta-analysis identified 11 RCTs that tested direct monetary incentives; all included studies measured *only direct effects*.[14(p418)] Similarly, a recent systematic review of all quantitative studies evaluating the impact of monetary incentives on COVID-19 vaccine intentions and uptake included 38 studies; all measured only direct effects.[15] Both the meta-analysis and the review found that financial incentives had small, positive, statistically significant impacts on increasing vaccination against COVID-19.[14(p418)] [15]

Prior research indicates that social spillover may occur in the context of vaccinations [9,16,17] and that spillover is not always positive.[9] We provide a more comprehensive test of a policy intervention very similar to that of Campos-Mercade et al. [12] but among the full population of a European city rather than a sample of paid survey panelists. Our study highlights the importance of measuring temporal and social spillover before a program is scaled up to a population.

Methods

Study Design

This address-cluster RCT was initiated in November 2021 and included all registered residents of the city of Ravensburg, Germany. The experimental design was pre-registered on OSF (DOI:10.17605/OSF.IO/2TRMB) and the clinical trial registry ISRCTN (DOI:10.1186/ISRCTN59503725). The Zeppelin University (Germany) ethics committee reviewed and approved this study on 29th October 2021.

Participants

The study population was drawn from the official Ravensburg city registry, which includes all adults who legally reside in the city. Three pre-registered exclusions, which totaled less than 1% of all residents, included a prison and an address that appeared to have 75 residents. All demographic information came from the city registry. For more details, please see SM, Section 9.

All vaccination events were conducted in strict adherence to the vaccination procedures stipulated by the German public health authorities, which include a mandatory consultation by a doctor prior to vaccination and the participant's signed confirmation of consent to get vaccinated. The Zeppelin University ethics committee waived participants' informed consent for inclusion in the study, the use of the city's administrative registry data, and the use of in-person observation to collect vaccination outcomes at the public vaccination events.

Randomization

10,032 addresses (41,548 residents) were randomized to two experimental conditions. After stratifying by cluster size and blocking within each strata (four addresses per block) to ensure balance, [18,19] all addresses were randomized into two groups (treatment and control). Then, one

resident in each address was randomly selected (a "cluster representative"). The cluster representatives that resided in treatment addresses would receive the treatment letter; all other individuals would receive the control letter. In other words, 5,016 individuals residing in 5,016 unique addresses were mailed a treatment letter. Their cohabitants and all individuals in the control addresses were mailed a control letter (36,532 individuals in all). Randomization code is available as part of the replication package (see SM, Section 7 for more details).

Procedures

The control and treatment letters were identical in every way, except that the treatment letter offered its recipients financial incentives for getting vaccinated against COVID-19. Mailed letters are the typical mode of communication between German government organizations and its residents; in 2021, 56% of all communication between citizens and public authorities in Germany was conducted via letter.[20] Both letters were sent out at the same time, with residents receiving their letters in the first week of November 2021. The letters informed recipients about seven public vaccination events that took place between November 13 and December 12, 2021. They were addressed from and signed by city officials and the Mayor of Ravensburg. At this time, the availability of COVID-19 vaccines and vaccination appointments for the general public in Germany was limited. The public vaccination events were intended to ameliorate this issue, especially as the start of the booster campaign created a considerable demand wave. The public events operated on a first-come first-served basis.

The treatment letter offered recipients a total of 40 Euros for vaccinating against COVID-19. Specifically, there were two financial incentives. The individual-level incentive (which was modeled after the financial incentive in [12]) offered recipients 20 Euros if they got vaccinated at one of the seven public vaccination events. To encourage *positive* social spillovers, we offered an additional community-level incentive: if the treatment letter recipient did get vaccinated, they could also receive an additional 20 Euros if more than 900 Ravensburg residents got vaccinated at the seven public vaccination events. The letter also specified that if the recipient was already fully vaccinated, they could still receive the community-level incentive if they transfer their treatment letter to another Ravensburg resident and this other individual presents the letter while getting vaccinated at one of the events. The full letters (in German and translated to English) are presented in SM Section 1.

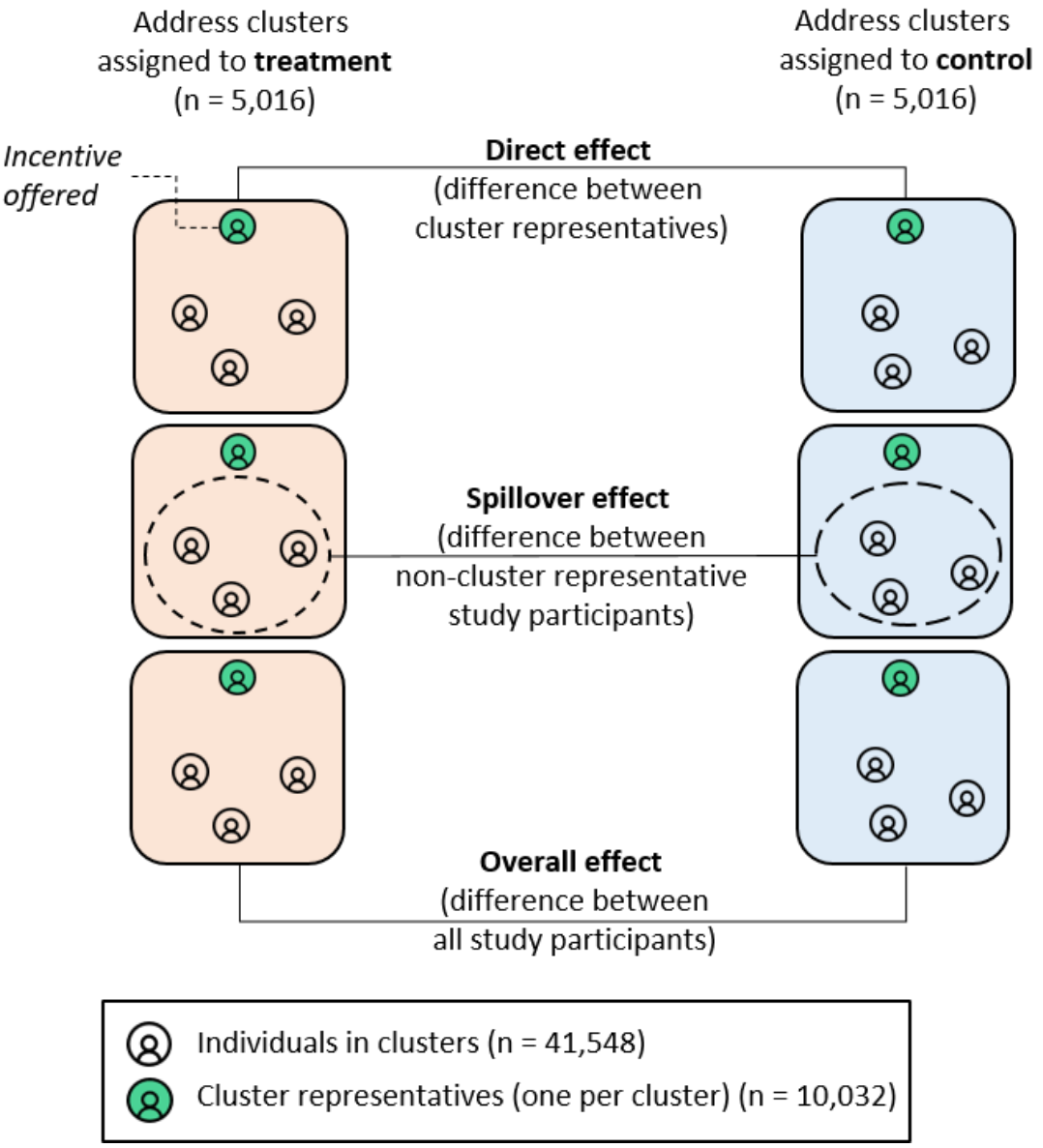
Outcomes

Our primary outcome of interest was vaccination uptake. Vaccination was recorded on-site during the events. Name and address information was linked to city administrative records and verified on-site. For more details on outcome data, please see SM Section 2.

Statistical Analysis

Statistical analysis was conducted as pre-registered. We estimated three types of commonly used[21] treatment effects: direct, spillover, and overall effects of monetary incentives (see Fig. 1). The direct effect compares the treated individuals in the treatment address-clusters with those individuals who would have been treated in the control address-clusters (i.e., the address-cluster representatives). The spillover effect compares the untreated individuals in treatment address-clusters with the individuals in the control address-clusters who were not randomly selected as address-cluster representatives. Finally, the overall effect compares all participants in the treatment address-clusters with those of the control address-clusters.

Fig. 1. Illustration of the estimated effects. The figure depicts randomization into treatment (left column) and control address-clusters (right column), the randomized selection of address-cluster representatives (green circles), and the estimated parameters of interest (direct, spillover, and overall effects).



As is standard practice [22,23], for direct treatment effects, we used covariate-adjusted (age, age^2, female, non-German) OLS regression with a set of indicators for randomization blocks and heteroskedasticity-robust standard errors; for spillover and overall treatment effects we clustered standard errors by address-clusters. Our results are robust to alternative specifications, including randomization inference [18] (see SM, Section 5.3). Our spillover analysis assumes that individuals living at the same address are in the same social network (for alternative social network operationalizations, see SM Section 5.1).

Results

As seen in the CONSORT diagram in Figure 2, the study universe consists of the entire adult (18 years or older) registered population of the city of Ravensburg (41,548 individuals in 10,032 addresses). All individuals were sent either a treatment letter or a control letter on 5 November, 2021. Seven public vaccination events were held as stated in the letters, between 13 November 2021 and 11 December 2021.

As seen in Table 1, residents were, on average, 49.96 years old (*SD* = 19.04), 51.33% were women, and 15.32% were foreign nationals. A total of 796 (1.92%) individuals were vaccinated at one of the seven announced events. The unadjusted difference-in-means and covariate-adjusted treatment effects are displayed in Figure 3 in a coefficient plot with 95% confidence intervals (for the full regression table, please see SM, Section 10.2, Table S40). The direct effect of receiving a monetary incentive letter on vaccination was directionally negative (-0.2 percentage points, n_{treat}=87, $n_{control}$=99, 95% CI [-0.77, 0.28]) though this was not statistically significant (p=.358). The spillover effect of the monetary incentive on vaccination is similarly a statistically insignificant -0.3 percentage points (n_{treat}=283, $n_{control}$=327, 95% CI [-0.60, 0.06], p=.111). The overall effect of being in an address-cluster that received the monetary incentive had a directional -0.3 percentage point statistically insignificant effect (n_{treat}=370, $n_{control}$=426, 95% CI [-0.55, 0.03], p=.076) on vaccinations.

Fig. 2. CONSORT diagram. The figure depicts the flow of participants through each stage of the randomized control trial.

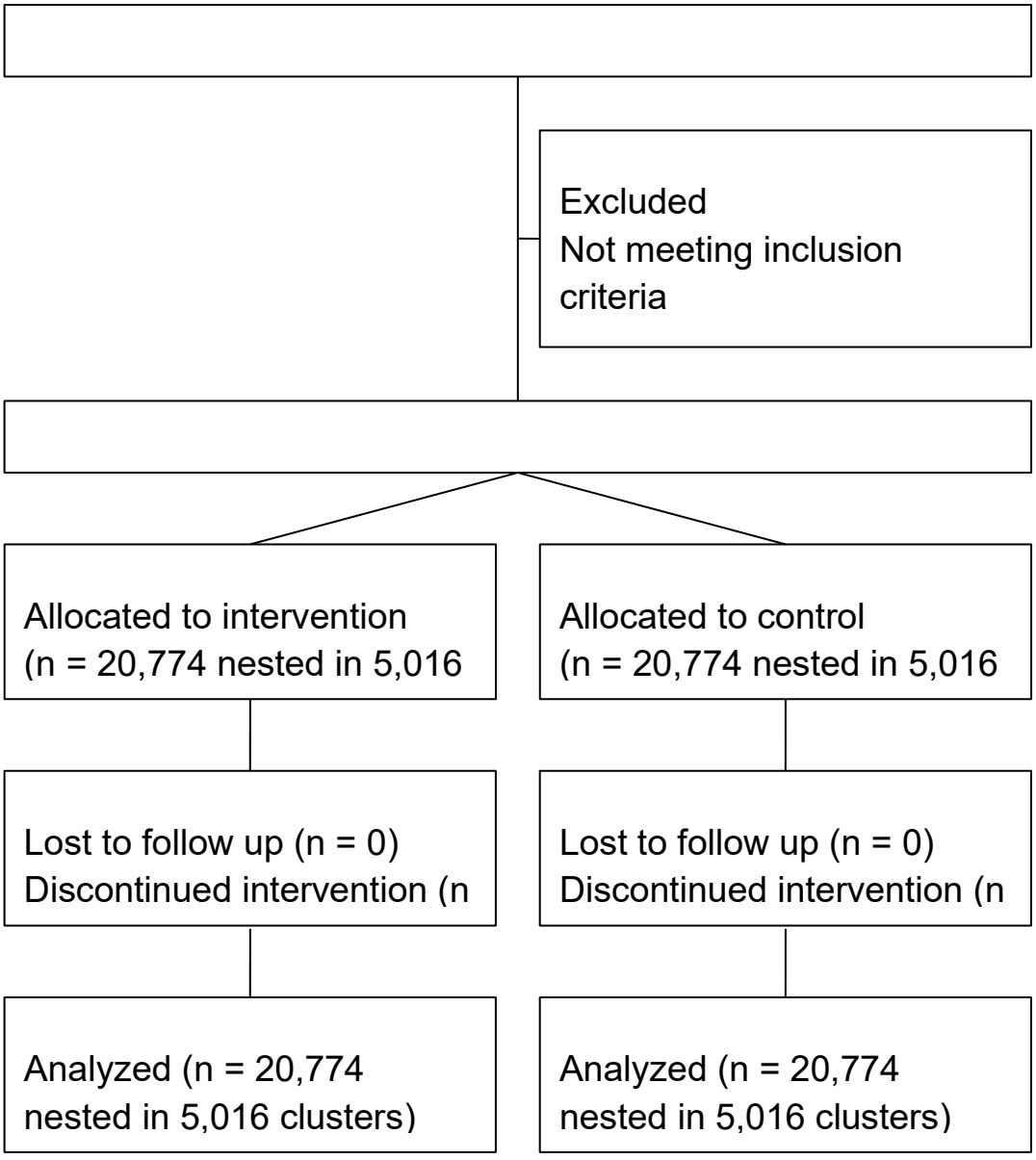
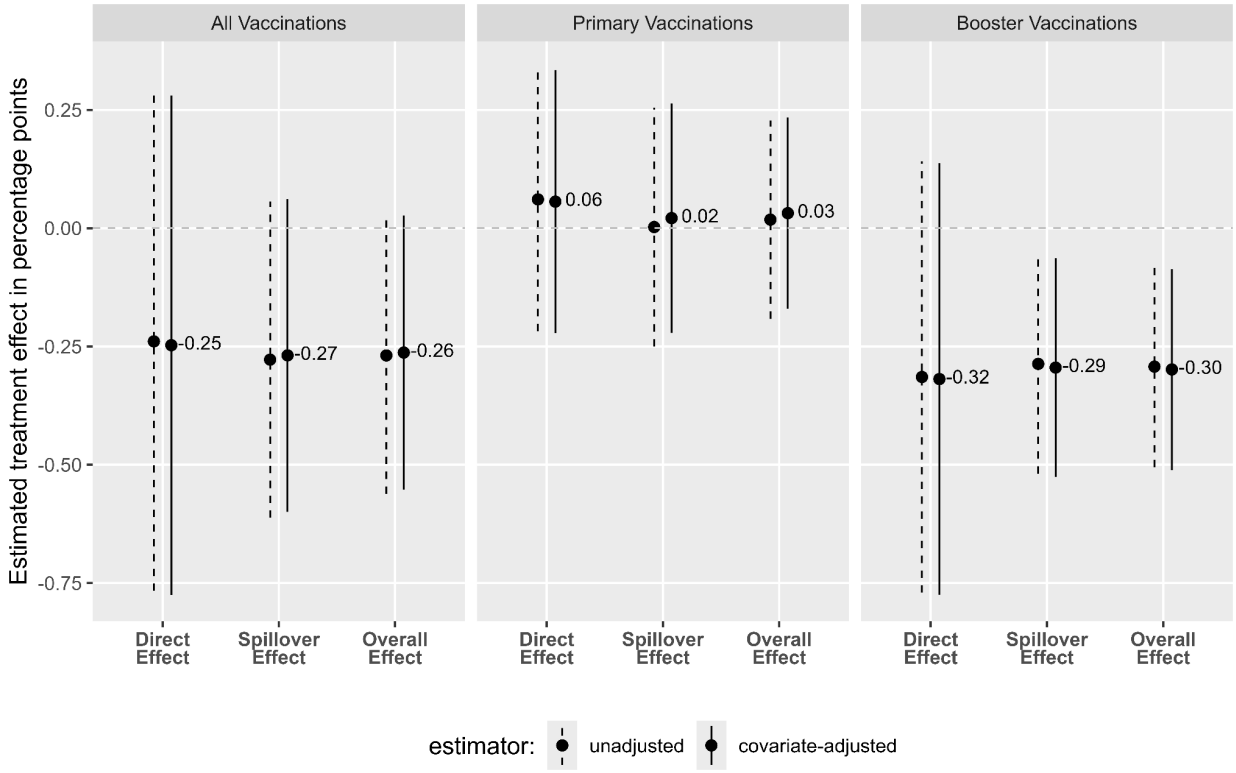


Table 1. Descriptives. Counts, percentages, means, and standard deviations for the study's treatment arms and the total population.

		Treatment arm		Control arm	
	Overall N = 41,548[1]	**Cluster representatives** N = 5,016[1]	**Co-habitants** N = 15,754[1]	**Cluster representatives** N = 5,016[1]	**Co-habitants** N = 15,762[1]
Mean Cluster Size	8.58 (8.30)	4.14 (4.28)	9.99 (8.71)	4.14 (4.30)	10.01 (8.79)
Age	49.96 (19.04)	53.22 (19.05)	49.03 (18.84)	53.06 (19.13)	48.85 (18.99)
Gender					
Female	21,326 (51.33%)	2,596 (51.75%)	8,010 (50.84%)	2,585 (51.54%)	8,135 (51.61%)
Male	20,222 (48.67%)	2,420 (48.25%)	7,744 (49.16%)	2,431 (48.46%)	7,627 (48.39%)
Foreign National					
Foreign National	6,367 (15.32%)	485 (9.67%)	2,623 (16.65%)	464 (9.25%)	2,795 (17.73%)
German	35,181 (84.68%)	4,531 (90.33%)	13,131 (83.35%)	4,552 (90.75%)	12,967 (82.27%)
Vaccinated at event					
Yes	796 (1.92%)	87 (1.73%)	283 (1.80%)	99 (1.97%)	327 (2.07%)
No	40,752 (98.08%)	4,929 (98.27%)	15,471 (98.20%)	4,917 (98.03%)	15,435 (97.93%)
Primary vaccination at event					
Yes	307 (0.74%)	27 (0.54%)	129 (0.82%)	23 (0.46%)	128 (0.81%)
No	41,241 (99.26%)	4,989 (99.46%)	15,625 (99.18%)	4,993 (99.54%)	15,634 (99.19%)
Booster vaccination at event					
Yes	489 (1.18%)	60 (1.20%)	154 (0.98%)	76 (1.52%)	199 (1.26%)
No	41,059 (98.82%)	4,956 (98.80%)	15,600 (99.02%)	4,940 (98.48%)	15,563 (98.74%)

[1] Mean (SD); n (%)

Fig. 3 Unadjusted and covariate-adjusted OLS regression estimates of the effects of the treatment on all vaccinations, primary vaccinations (first and second dose), and booster vaccinations. Models include a set of randomization-block indicators as covariates. Error bars denote 95% confidence intervals.



Additionally, we pre-registered analyses of the effects of the monetary incentives on primary vaccinations (first and second doses) and boosters. The middle and left panel in Figure 3 present the results for primary vaccinations and booster vaccinations separately (for full regression tables, please see SM, Section 10.2, Table S43). Impacts on first doses were not meaningfully different from that of primary doses (see SM, Table S45).

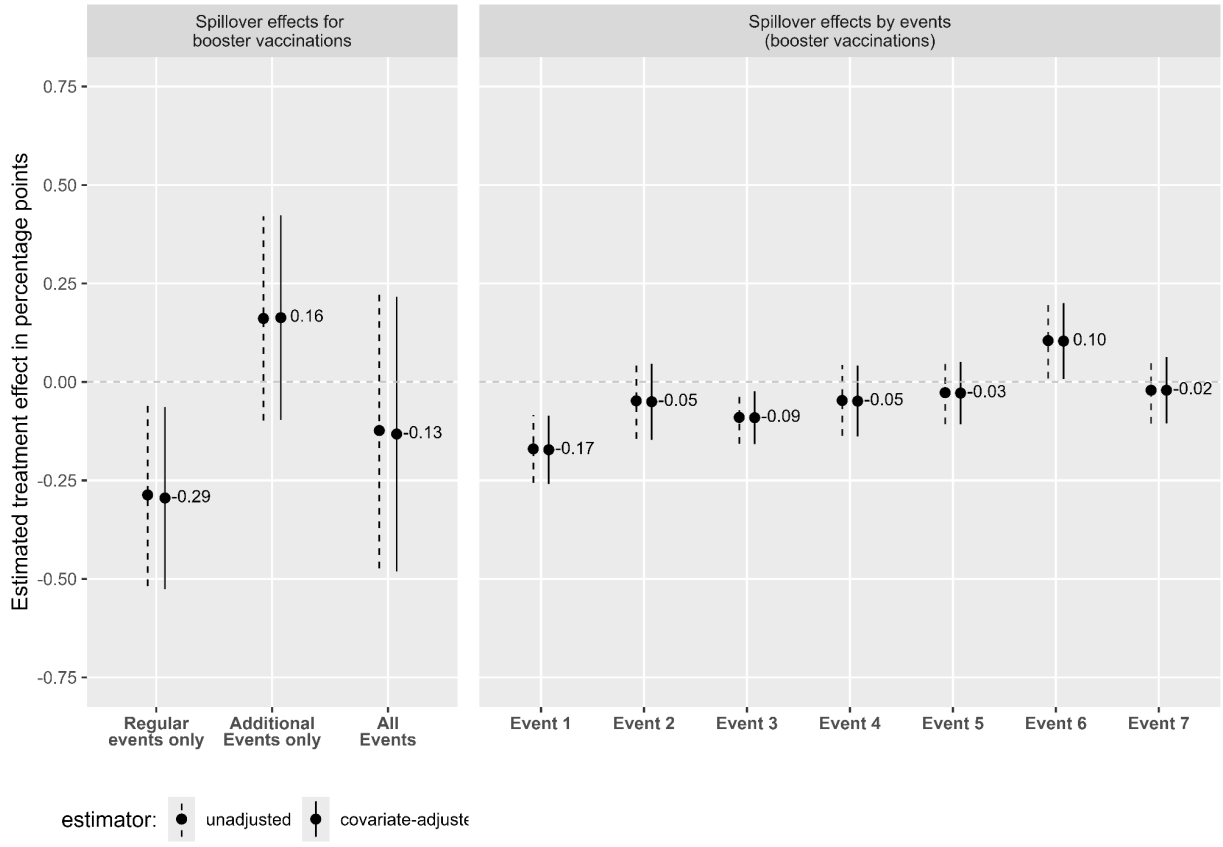
We find a negative statistically significant spillover effect of -0.3 percentage points (n_{treat}=154, $n_{control}$=199, 95% CI [-0.5, -0.1], p=.01) for booster vaccinations. In other words, untreated cohabitants in a treatment address-cluster are less likely to get vaccinated at publicly announced vaccination events as compared to their counterparts in a control address-cluster. The overall effect is also negative (-0.3 percentage points) and statistically significant (n_{treat}=214, $n_{control}$=275, 95% CI [-0.5, -0.1], p<.01): individuals in treatment address-clusters are less likely to receive a booster vaccination at the announced vaccination events, as compared to residents in

the control address-clusters. We found that this spillover backlash on boosters was substantially larger in smaller households (-0.7 percentage points, n_{treat}=109, $n_{control}$=147, 95% CI [-1.2, -0.3], p<.01), which further supports the interpretation that this is a socially-mediated backlash.

After the first vaccination event, it became clear that vaccination demand exceeded initial predictions and the city decided to organize 11 additional vaccination events. These additional events were not mentioned in either the treatment or control letters and occurred between 1 December and 18 December. We updated our preregistration plan when the possibility of additional events was made known to us by the city (12 November 2021). Since we collected outcome data from these additional events, this policy change gave us a unique opportunity to supplement our analysis of temporal spillover effects with outcome data that would normally be censored from a research team.

707 residents were vaccinated at these additional events. As part of the pre-registered exploratory analysis, we also examine the impact of financial incentives on vaccination at additional events. As seen in the left panel in Figure 4, when we limited our analysis only to vaccinations at these additional events, we see a directionally positive spillover effect on booster vaccinations of a similar magnitude to the backlash (+0.2 percentage points, n_{treat}=224, $n_{control}$=200, 95% CI [-0.1, 0.4]) though this effect was not statistically significant (*p*=.22). When we re-run our main analysis with vaccinations at additional events as affirmative vaccinations, the negative spillover backlash on boosters disappears (-0.1 percentage points, n_{treat}=378, $n_{control}$=399, 95% CI [-0.5, 0.2], p=.46). In other words, what appeared to be a social backlash when we used censored data may be a displacement effect: individuals who lived with someone who received a monetary incentive may have postponed their booster vaccination to the unannounced events. There is additional suggestive evidence for this interpretation. The first three events had long waiting lines and when we examined treatment effects by event (as pre-registered), we found that the statistically significant negative backlash was localized entirely in the first event (see right panel in Figure 4).

Fig. 4 Unadjusted and covariate-adjusted OLS regression estimates of the spillover effects of the treatment on vaccinations at regular, additional, and all events (left panel) and spillover effects by events (right panel). Models include a set of randomization-block indicators as covariates. Error bars denote 95% confidence intervals.



Discussion

Consistent with recent results [24,25], a city-wide monetary incentive that was randomly allocated to individual households did not significantly increase people's propensity to get vaccinated. However, when examining only booster vaccinations, a small negative spillover effect was found, which we believe is caused by a displacement of vaccinations from advertised events to future unadvertised events. Our results illustrate the presence of spillovers and the related pitfalls of missing data when evaluating the effectiveness of policy interventions. A random (representative) sample may produce accurate estimates of direct effects, but most random samples fail to capture spillover effects, for instance, within a household.

Missingness of outcome data played an important role in interpreting this unexpected result. Since the city decided to host additional vaccination events that were not mentioned in the letters that were sent to the participants in our study, we were able to get insight into possible displacement effects. Though we saw negative spillover effects for vaccination during advertised events, we also saw positive (though non-significant) spillover effects on vaccinations during unadvertised events. This result complemented the fact that the first few advertised events had long lines and some latecomers were turned away due to the limited number of vaccine doses. This may mean that individuals living at the same address as someone who received a monetary incentive may have assumed that most of the attendees of the advertised events were there for the monetary incentive. And so, to avoid long lines, they may have postponed vaccination.

Our results also illustrate the limits of moderate monetary incentives as a public health intervention. Individual-level incentives are most effective when the individual intrinsic benefit of getting vaccinated is only slightly lower than the individual cost of vaccination. However, with the rapid surge in infections and hospitalizations in November 2021, we found long lines during the initial events; nearly half of all attendees of the events were from outside the city, and most vaccinations during the events were boosters. This may indicate that, for the average city resident, the benefit of getting vaccinated may have already been far greater than the cost. The individual-level incentive may have simply been unnecessary for those who already intended to be vaccinated. Conversely, the lack of effect of the individual-level incentive on first doses may mean that the vaccine-hesitant required much higher compensation than what was offered. There were similar takeaways for the community-level incentive, though the community-level incentive may have been too complicated to have widespread positive spillovers. Specifically, the community-level incentive had multiple pathways of operation. First, it acted to complement the individual incentive for recipients who were not yet fully vaccinated. Each recipient of the community-level incentive was urged to encourage others to get vaccinated in order to receive an additional monetary reward. However, the community-level goal may have been perceived as too high. Nevertheless, the goal of 900 vaccinations was ultimately reached, doubling the monetary incentive people have received.

More broadly, our results illustrate the advantage of research designs that measure social and temporal spillover. Our RCT design was inspired by the promising results in Campos-Mercade et al. [12], but as we illustrate in our population-level RCT, it is not enough to scale a

public health policy based on the results of an RCT conducted over a random sample of beneficiaries. If we measured only direct effects, we would see that, unlike Campos-Mercade et al. [12], monetary incentives had a null effect on the population level–despite strong similarities in the monetary incentive interventions across the two studies. That, in itself, is informative, but concluding that monetary incentives, at the very least, did no harm in Ravensburg is an incorrect conclusion. By examining both temporal and social spillover, we identified a small but statistically significant backlash, which illustrated that a seemingly benign intervention could still lead to adverse outcomes (even if it means, for instance, delaying when an individual gets a booster shot). Social and temporal spillover from other health policy interventions could foreseeably have larger, more serious consequences.

This study illustrates the importance of measuring spillovers from health policy interventions to identify potential social backlashes before the policy is rolled out on a much larger scale. In this case, the postponement of booster COVID-19 vaccination among partially vaccinated persons may not have had a clinically significant backlash. However, this should not imply that social spillovers of paying for COVID-19 vaccination could have safely been ignored. First, we believe the postponement effect reported here is a conservative estimate of spillover backlash to individual-level monetary incentives. Since there was compelling pre-existing evidence of the possibility of negative social spillover to individual-level monetary incentives [23], we incorporated community-level incentives in the treatment instead of just replicating Campos-Mercade et al. [12] on a population level. There is no reason to believe that this would attenuate the impact of individual-level incentives, but it could ameliorate the possibility of backlash. We are also only able to observe both a limited time period (i.e., we do not know if the social spillover impacts the affected individuals' vaccination decisions years later) and scope (i.e., we are able to observe only social spillover among cohabitants and not other members of an incentive recipient's social network). The fact that we see *any* backlash from the combination of individual-level and community-level incentives should be viewed as a cautionary note for the use of monetary incentives in this context. Additionally, other public health contexts that require immediate community action (such as in the early stages of a burgeoning epidemic) may mean that postponement of a health behavior could have meaningful consequences for containment efforts [26,27]. As such, we strongly recommend incorporating measures of spillover into RCTs before a policy is implemented at scale.

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Notes

The Zeppelin University ethics committee reviewed and approved this study on 29th October 2022. All vaccination events were conducted in strict adherence to the vaccination procedures stipulated by the German public health authorities, which includes a mandatory consultation by a doctor prior to vaccination and the participant's signed confirmation of consent to get vaccinated. The Zeppelin University ethics committee waived participants' informed consent as it relates to being included in the study and the collection of data as to whether or not a participant was observed consenting to be vaccinated at the public event. The study was performed in accordance with relevant guidelines and regulations.

The experimental protocol was pre-registered on OSF (DOI:10.17605/OSF.IO/2TRMB) and the clinical trial registry ISRCTN (DOI:10.1186/ISRCTN59503725).

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Investigation: JT, SJ, FK, DV

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DV and FK supervised outcome data collection. DV conducted the data analysis and JT verified all results using the de-identified dataset.

Competing interests

Authors declare that they have no competing interests.

Data and materials availability

A de-identified dataset is available at OSF (https://osf.io/uh2sd/).

Supplementary Materials for

Spillover Effects of Financial Incentives for COVID-19 Vaccination: A Population-level Cluster Randomized Controlled Trial

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1 Section 1: Treatment and Control Letters

1.1 Control Letter (English Translation)

[Personalized Address]

Get your COVID-19 vaccination now!

Hello [ACADEMIC TITLE] [NAME],

We are contacting you today for your coronavirus vaccination.

Are you not yet fully vaccinated? Then please go for vaccination on one of the following dates:

- **13.11.:** Saturday, 9 a.m.-2 p.m.
- **19-20.11.:** Friday, 3-9 p.m. or Saturday, 9 a.m.-2 p.m.
- **26-27.11.:** Friday, 3-9 p.m. or Saturday, 9 a.m.-2 p.m.
- **10-11.12.:** Friday, 3-9 p.m. or Saturday, 9 a.m.-2 p.m.

Place: City Hall, Marienplatz 26, 88212 Ravensburg **(without appointment)**

Are you already fully vaccinated? Then motivate your environment from Ravensburg now to a vaccination on one of the mentioned dates.

Available are vaccines from BioNTech/Pfizer, Moderna and Johnson & Johnson. Please bring, if available, vaccination certificate, identity card and insurance card of the health insurance. For more information, please visit the following website: [INDIVIDUAL LINK].

Kind regards

[Signature, name, and picture of the local Mayor, chairman of the local association of the medical profession, leading physician of the local public clinic]

[Text on the right side in the column: Contact details of the sender, i.e. the Mayor, date, reference number, phone number and email address for questions, text: "all info via the link or directly via the QR code", translation note for the languages "English, Spanish, Russian, Arabic, Turkish", additional note: "This vaccination campaign is scientifically accompanied. It is supported by:", logos of sponsors]

1.2 Treatment Letter (English Translation)

[Personalized Address]

Get your COVID-19 vaccination now, receive a 20 Euro voucher and secure the chance to win another 20 Euro!

Hello [ACADEMIC TITLE] [NAME],

Congratulations! The city of Ravensburg is giving away shopping vouchers. You are one of 5,000 **randomly selected** people from Ravensburg who can receive these limited vouchers.

Then please go for vaccination on one of the following dates.

Are you not yet fully vaccinated? Then please go for vaccination on one of the following dates. **You will receive your 20 Euros voucher for this** by mail after the vaccination.

- **13.11.:** Saturday, 9 a.m.-2 p.m.
- **19-20.11.:** Friday, 3-9 p.m. or Saturday, 9 a.m.-2 p.m.
- **26-27.11.:** Friday, 3-9 p.m. or Saturday, 9 a.m.-2 p.m.
- **10-11.12.:** Friday, 3-9 p.m. or Saturday, 9 a.m.-2 p.m.

Place: City Hall, Marienplatz 26, 88212 Ravensburg **(without appointment)**

After your vaccination on one of these dates, you also have a **chance to win**: if more than 900 people from Ravensburg get vaccinated on these dates, you will receive **your 2nd voucher worth 20 Euros** by mail.

Are you already fully vaccinated? Then motivate your environment from Ravensburg now to a vaccination on one of the mentioned dates.If this person shows your letter at one of the mentioned dates, you secure your **chance to win**: If more than 900 people from Ravensburg get vaccinated, you will receive **your 20 Euros voucher** by mail.

Available are vaccines from BioNTech/Pfizer, Moderna and Johnson & Johnson. Please bring, if available, vaccination certificate, identity card and insurance card of the health insurance. For more information, please visit the following website: [INDIVIDUAL LINK].

Kind regards

[The rest of the text is identical to the control letter.]

1.3 Control Letter (Original)

Jetzt Ihre COVID-19 Impfung bekommen!

Stadt Ravensburg

Stadtverwaltung · Postfach 21 80 · 88191 Ravensburg
Herrn
[redacted]

Oberbürgermeister
Dr. Daniel Rapp
Lederhaus
Marienplatz 26
88212 Ravensburg
www.ravensburg.de

5.11.2021
Zeichen: TEST2

Für Rückfragen:
✆ 0751 82-780
✉ mail@ravensburg.de

Alle Infos über den Link oder direkt über den QR-Code:

Jetzt Ihre COVID-19 Impfung bekommen!

Guten Tag Herr [redacted],

wir kontaktieren Sie heute zur Schutzimpfung gegen das Coronavirus.

Sie sind noch nicht vollständig geimpft? Dann gehen Sie bitte an einem der folgenden Termine zur Impfung:

- **13.11.:** Samstag, 9–14 Uhr
- **19.–20.11.:** Freitag, 15–21 Uhr oder Samstag, 9–14 Uhr
- **26.–27.11.:** Freitag, 15–21 Uhr oder Samstag, 9–14 Uhr
- **10.–11.12.:** Freitag, 15–21 Uhr oder Samstag, 9–14 Uhr

Ort: Rathaus, Marienplatz 26, 88212 Ravensburg **(ohne Anmeldung)**

Sie sind bereits vollständig geimpft? Dann motivieren Sie jetzt Ihr Umfeld aus Ravensburg zu einer Impfung an einem der genannten Termine.

Verfügbar sind die Impfstoffe von BioNTech/Pfizer, Moderna und Johnson & Johnson. Bitte bringen Sie, falls vorhanden, Impfpass, Personalausweis und Versichertenkarte der Krankenkasse mit. Mehr Informationen erhalten Sie auf folgender Webseite: **https://impfkampagne.ravensburg.de/TEST2**

The website offers all information in other languages: English, Español, Русский, العربية, Türkçe

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MVZ Labor Ravensburg
Labor Dr. Gärtner

wifo PRO RAVENSBURG

Freundliche Grüße



Dr. Daniel Rapp
Oberbürgermeister der Stadt Ravensburg



Dr. med. Hans Bürger
Facharzt für Allgemeinmedizin, Vorsitzender der Ärzteschaft des Landkreises Ravensburg



PD Dr. med. Andreas Straub
Chefarzt der Klinik für Anästhesie, Intensiv-, Notfall- und Schmerzmedizin, Oberschwabenklinik gGmbH

1.4 Treatment Letter (Original)

Jetzt Ihre COVID-19 Impfung bekommen und 20 Euro-Gutschein sichern!

Stadt Ravensburg

Stadtverwaltung · Postfach 21 80 · 88191 Ravensburg
Herrn
[redacted]

Oberbürgermeister
Dr. Daniel Rapp
Lederhaus
Marienplatz 26
88212 Ravensburg
www.ravensburg.de

5.11.2021
Zeichen: TEST1

Für Rückfragen:
✆ 0751 82-780
✉ mail@ravensburg.de

Alle Infos über den Link oder direkt über den QR-Code:

The website offers all information in other languages: English, Español, Русский, العربية, Türkçe

Diese Impfaktion wird wissenschaftlich begleitet. Sie wird unterstützt von:

GnW Gemeinsam neue Wege

MVZ Labor Ravensburg
Labor Dr. Gärtner

wifo PRO RAVENSBURG

Jetzt Ihre COVID-19 Impfung bekommen, einen 20 Euro-Gutschein erhalten und die Gewinnchance auf weitere 20 Euro sichern!

Guten Tag Herr [redacted]

Herzlichen Glückwunsch! Die Stadt Ravensburg verlost Einkaufsgutscheine. Sie gehören zu 5.000 **zufällig** ausgewählten Personen aus Ravensburg, die diese limitierten Gutscheine erhalten können.

Sie sind noch nicht vollständig geimpft? Dann gehen Sie bitte an einem der folgenden Termine zur Impfung. **Sie erhalten dafür nach der Impfung Ihren 20 Euro-Gutschein** per Post.

- **13.11.:** Samstag, 9–14 Uhr
- **19.–20.11.:** Freitag, 15–21 Uhr oder Samstag, 9–14 Uhr
- **26.–27.11.:** Freitag, 15–21 Uhr oder Samstag, 9–14 Uhr
- **10.–11.12.:** Freitag, 15–21 Uhr oder Samstag, 9–14 Uhr

Ort: Rathaus, Marienplatz 26, 88212 Ravensburg **(ohne Anmeldung)**

Nach Ihrer Impfung an einem dieser Termine haben Sie zudem eine **Gewinnchance**: Wenn sich mehr als 900 Personen aus Ravensburg an diesen Terminen impfen lassen, erhalten Sie **Ihren 2. Gutschein im Wert von 20 Euro** per Post.

Sie sind bereits vollständig geimpft? Dann motivieren Sie jetzt eine erwachsene Person aus Ravensburg zur Impfung. Wenn diese Person Ihr Schreiben bei einem der genannten Termine vorzeigt, sichern Sie sich die **Gewinnchance**: Wenn sich mehr als 900 Personen aus Ravensburg impfen lassen, dann erhalten Sie **Ihren 20 Euro-Gutschein** per Post.

Verfügbar sind die Impfstoffe von BioNTech/Pfizer, Moderna und Johnson & Johnson. Bitte bringen Sie, falls vorhanden, Impfpass, Personalausweis und Versichertenkarte der Krankenkasse mit. Mehr Informationen erhalten Sie auf folgender Webseite: **https://impfkampagne.ravensburg.de/TEST1**

Freundliche Grüße



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Oberbürgermeister der Stadt Ravensburg



Dr. med. Hans Bürger
Facharzt für Allgemeinmedizin, Vorsitzender der Ärzteschaft des Landkreises Ravensburg



PD Dr. med. Andreas Straub
Chefarzt der Klinik für Anästhesie, Intensiv-, Notfall- und Schmerzmedizin, Oberschwabenklinik gGmbH

2 Section 2: Data Collection

This section describes how our outcome variables were collected.

The variable "**information uptake"** describes whether a letter recipient accessed the personalized link or QR code that was prominently displayed on both the control and treatment letters.[1] The links led to a city website, which informed residents about COVID-19 vaccines and the upcoming city-organized vaccination events, and encouraged them to get vaccinated. As noted in the letter, the website could be viewed in German and–as an additional service for migrant residents–in other languages commonly spoken in the region (English, Spanish, Russian, Arabic, and Turkish).

The link addresses incorporated a randomly-generated five-character code (e.g., 123AB) to identify individual click-behavior (i.e., which recipient visited the website and which did not). For ease of access and to enable us to measure link clicks in accordance with EU GDPR regulations, the links shortening web application "rebrand.ly" and its secure API were used to create the personalized links (i.e., link.com/123AB) and generate the "unique clicks" variable.

The variable **"vaccination uptake"** describes whether a resident was vaccinated at one of the advertised public vaccination events. This data was collected as follows.

The persons who arrived at the vaccination site to be vaccinated were asked to wait in line outside of the site in order to reduce risk of infection. City security staff was present on-site. Initially, security staff invited the first person in line into the vaccination site and progressed through the line in this manner until the vaccine doses were depleted for the day. Once vaccine doses were depleted the remaining individuals in the line were asked to return on a different day. Since this approach risked disincentivizing vaccination, a waitlist system was introduced starting with the fourth announced event (11/26/2021). Under the waitlist system, the city distributed numbered paper cards, with the total number of cards determined by the number available vaccine doses. The numbers on the cards were associated with certain vaccination appointment times, so that waitlisted people could come back to the vaccination site instead of waiting in long lines outside of the site. This had the benefit of reducing wait time and the risk of infection.

The process inside the vaccination site building was the same with and without the waitlist system. Once inside, the individual to be vaccinated proceeded to the front desk where city officials collected data. The individual was asked to present an ID or comparable document. Officials at the front desk would copy this information into a spreadsheet. Then, the individual was directed to a different table (in the same physical space), where medical staff would provide a medical consultation as required by law. In other words, the medical staff would provide information about the type of vaccination that would be administered and answer individual health questions. If the individual decided to opt out after the medical consultation, the data collection team was informed and made a note of this in the collected data. There were four opt-outs; all four were coded as unvaccinated in all analyses in the study. Finally, the vaccination was administered and individuals were supervised afterward in a monitoring room in case of adverse (i.e., allergic) reactions. The entirety of this data collection process was overseen by one of the researchers at all events. The researchers then matched the collected data with the list of letter recipients.

[1] We tracked both whether or not a personalized link/QR code was accessed *and* how often it was accessed.

3 Section 3: Timeline of Experiment and Relevant Policy Changes

This population-level field experiment was conducted in the city of Ravensburg, Germany in the fall of 2021. Details of the policy context at the time the experiment was conducted can be found in the SM (Sections 1.1 and 1.10) of a related paper, Jilke et al. (2023). We provide an extended version of the timeline in Jilke et al. (2023, SM, p. 4) here:

10/29/2021	The Zeppelin University ethics committee issues its approval of this study.
10/29/2021	All Ravensburg residents are randomized to be sent either a treatment letter (with information about the city's upcoming free, public vaccination events AND individual-level and community-level financial incentives) or a control letter (identical to the treatment letter but with no individual-level or community-level financial incentives).
11/05/2021	Treatment and control letters are sent out.
11/05/2021	The federal health ministry and the state ministries in Germany recommend booster vaccination for all persons aged 18 and over.
11/10/2021	Federal and state authorities decide to offer only the Moderna vaccine to persons aged 30 and older, allotting the Pfizer/BioNTech vaccine for younger persons. (There were very few doses of the Pfizer/BioNTech vaccine available in Germany, at this time.)
11/13/2021	The first of the seven announced vaccination events takes place.
11/15/2021	Due to high demand and long lines at the first vaccination event, Ravensburg organizes 11 additional vaccination events. There is no additional mailed communication about these events, however the original letters included a QR code to a government website and a hotline number. From this point on, the website and hotline included information about the additional, unannounced events.
11/17/2021	The regional health ministry of Baden-Wuerttemberg introduces the "2G rule," restricting access to most shops and restaurants to people who either 1) received the first and second dose or 2) recovered from COVID-19.
11/18/2021	The Standing Committee on Vaccination, an independent expert committee in Germany, recommends booster vaccination for all persons older than 18, six months after the second dose.
11/19/2021	The second of the seven announced vaccination events takes place.
11/20/2021	The third of the seven announced vaccination events takes place.
11/22/2021	The federal health ministry restricts the delivery of the BioNtech/Pfizer vaccine to reserve it for minors under the age of 18 and offers Moderna vaccines as an alternative.
11/26/2021	The fourth of the seven announced vaccination events takes place. Waitlist system first introduced.
11/26/2021	The German air force starts to transport patients to other parts of the country and neighboring countries due to overcrowded hospitals.
11/27/2021	The fifth of the seven announced vaccination events takes place.
11/27/2021	The federal health ministry officially confirms the first cases of Omicron in Germany and further restricts travel from other countries.

11/30/2021	Chancellor Scholz and the heads of German state governments announce that the 2G rule has become the nationwide standard and confirm plans for a vaccination mandate.
12/01/2021	The first of the additional (unannounced) vaccination events takes place.
12/04/2021	The regional health ministry of Baden-Wuerttemberg introduces the "2G+ rule", restricting access to most shops and restaurants to people who either 1) received the first and second dose or 2) recovered from COVID-19 and can provide a recent negative test.
12/08/2021	The state of Baden-Wuerttemberg opens further vaccination centers exclusively for booster vaccinations.
12/08/2021	The second of the additional (unannounced) vaccination events takes place.
12/09/2021	The third of the additional (unannounced) vaccination events takes place.
12/10/2021	The sixth of the seven announced vaccination events takes place.
12/11/2021	The seventh of the seven announced vaccination events takes place.
12/15/2021	The fourth of the additional (unannounced) vaccination events takes place.
12/16/2021	The fifth of the additional (unannounced) vaccination events takes place.
12/17/2021	The sixth of the additional (unannounced) vaccination events takes place.
12/18/2021	The seventh of the additional (unannounced) vaccination events takes place.
12/21/2021	The national government and the heads of German state governments announce stricter social restrictions. Starting on 12/28/2021, private gatherings of vaccinated and recovered persons are restricted to a maximum of ten persons; private gatherings of unvaccinated persons are limited to the person's own household and a maximum of two persons from another household. The Standing Committee on Vaccination recommends booster vaccinations for all persons older than 18 years of age, three months after the second dose.
12/22/2021	The eighth of the additional (unannounced) vaccination events takes place.
12/23/2021	The ninth of the additional (unannounced) vaccination events takes place.
12/29/2021	The tenth of the additional (unannounced) vaccination events takes place.
12/30/2021	The eleventh of the additional (unannounced) vaccination events takes place.

4 Section 4: Investigating Postponement Spillovers

Field studies looking at vaccination outcomes must consider censored data. Like most countries,[2] Germany does not have a national vaccination registry, so it is not possible to observe whether an individual was vaccinated outside of the city-organized events. Our study was initially expected to focus on only the public events announced in the letters sent out by the city. However, when city leadership decided to organize additional events, this presented us with a unique opportunity to investigate the possibility that direct or indirect exposure to the monetary incentives may make a participant more likely to get vaccinated at other venues.[3] In other words, we are able to observe outcome data that may normally be censored.

Table S1: Treatment effects by type of vaccination considering only additional vaccination events. Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations. Effects are estimated for primary vaccinations (first and second dose) and booster vaccinations separately.

	Primary vaccination			**Booster vaccination**		
	(1) Direct Effect	(2) Spillover Effect	(3) Overall Effect	(4) Direct Effect	(5) Spillover Effect	(6) Overall Effect
Treatment	0.195+	-0.083	-0.015	0.033	0.161	0.133
	(0.115)	(0.078)	(0.066)	(0.215)	(0.132)	(0.116)
Age	-0.003	-0.003	-0.005	0.044	-0.009	-0.001
	(0.022)	(0.012)	(0.010)	(0.034)	(0.021)	(0.018)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.015	0.044	0.023	0.036	0.156	0.106
	(0.147)	(0.075)	(0.060)	(0.250)	(0.129)	(0.103)
Non-German	1.089**	1.000***	0.998***	-0.241	-0.255	-0.303+
	(0.347)	(0.178)	(0.161)	(0.417)	(0.207)	(0.182)
Std. Errors	Hetero-skedasticity-robust	by address cluster	by address cluster	Hetero-skedasticity-robust	by address cluster	by address cluster
n	9,736	30,479	40,215	9,814	30,763	40,577

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

In Table S1, we present our main analyses only for the additional events not advertised in the treatment and control letters. We are underpowered in this analysis but see suggestive evidence that the backlashes we report in the main text may be at least partially explained by displacement. Specifically, we observe a positive spillover effect for boosters of 0.2 percentage points (though it is

[2]Few countries have national vaccination registries (Papagiannis et al., 2022) and the systems that do exist have fared poorly in data quality audits - "10–60% of immunization records lack important information or contain errors" (Atkinson et al., 2020).

[3]Our pre-registration plan was updated as soon as the possibility of additional events was made known to us (11/12/2021).

not statistically significant). Given that our main analysis found a negative booster spillover effect of similar magnitude in the opposite direction (see Table 3 of the main text), this is suggestive evidence that individuals in the same household as someone who received a financial incentive may have postponed vaccination to a future event that was not announced in the treatment letter.

There is additional circumstantial evidence that comports with this interpretation. The first three announced events had long lines and there were cases of people being turned away due to insufficient vaccine doses. Additionally, 55.9% of attendees at the advertised events were from outside of the city of Ravensburg, which would potentially crowd out impacts of our treatment, as these individuals were never randomized. An individual residing in the same household as someone who received a monetary incentive may feel that long lines at the advertised events are indicative of people wanting to claim their monetary incentive. As such, these cohabitants of treatment letter recipients may seek to get vaccinated at other events.

The additional events are not the only alternative venues where a Ravensburg resident may seek out vaccination. Still, at the time of this study, vaccine doses were of extremely limited supply in Germany and appointments were hard to come by. To get some sense as to the availability of other vaccination options, we asked all those who attended the public vaccination events whether their physician offers COVID-19 vaccinations. Because this survey conditions on attending the public vaccination events (i.e., survey respondents are systematically different from people who didn't attend the public vaccination events), these data are likely to have high-levels of self-selection and should be viewed only as a rough benchmark of our participants' alternative vaccination options. From Table S2, we see that few participants in the survey viewed vaccination at their GP's office as a viable alternative.

Table S2: Potential displacement effects. Participants were asked whether they already planned to get vaccinated by a GP in the following weeks

Answer	Total
No or 'I do not have a GP'	272 (34.17%)
Yes	129 (16.21%)
GP has no appointments left	263 (33.04%)
GP does not vaccinate	88 (11.06%)
No answer	44 (5.53%)

To investigate whether long lines had any differential impact on a treatment household and control household members' decision to wait in line to get vaccinated, we present treatment and control vaccinations by day and time of day in Fig. S1-S2. The results of these analyses are not clear, but there does seem to be some association between experimental conditions and the time of day the individual was vaccinated.

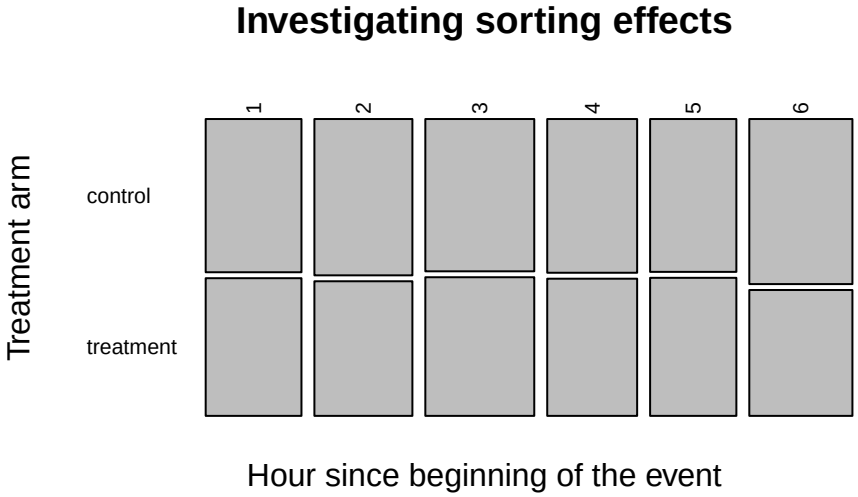


Figure S1: Distribution of vaccinations administered by treatment arm and hours since the beginning of the event. Figure considers all regular events.

We note that the first three days had long lines and there was more demand than supply of vaccine doses. The remaining days had expanded supplies of vaccines and used a waitlist. The waitlist system is described in detail above, in Section 2, Data Collection. We find that 78.6% of participants in treatment households got vaccinated after being on the waitlist compared to only 45.0% in control households (p=.109).

While the sum of this evidence is far from strong, it does provide some support for the theory that long lines may have had an effect on how the monetary incentive directly and indirectly affected vaccination behaviors.

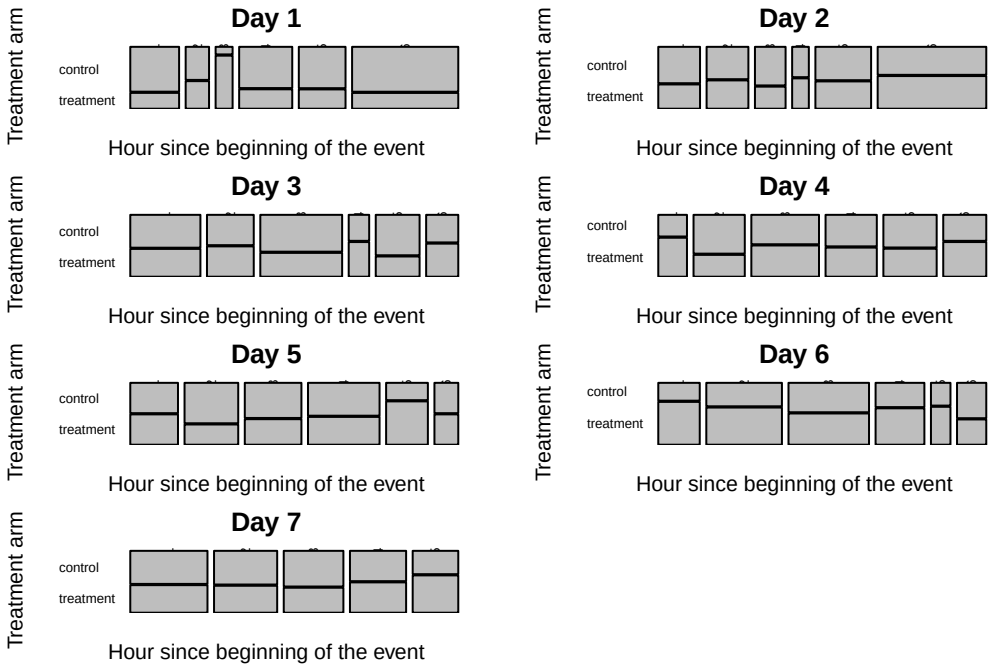


Figure S2: Daily distribution of vaccinations administered by treatment arm and hours since the beginning of the event.

5 Section 5: Robustness Checks

5.1 Alternative Cluster Specifications

Registry address data does not include apartment numbers and, for some larger address-clusters, potential outcomes may not be correlated among residents of the same address. To address this possibility, we pre-registered an analysis that reformulated clusters to a shared address and a shared last name.[4] As seen in Tables S3-S8, this specification yields results that are broadly in line with our main analyses.

5.1.1 Same Name and Address as Clusters

Table S3: Main results with clustered standard errors defined as shared name and address. Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations and information uptake.

	(1) Spillover effect Vaccinations	(2) Overall effect Clicks	(3) Overall effect Vaccinations
Treatment	-0.269+	0.000	-0.263+
	(0.162)	(0.147)	(0.143)
Age	0.014	0.140***	0.017
	(0.023)	(0.021)	(0.020)
Age^2	0.000	-0.001***	0.000
	(0.000)	(0.000)	(0.000)
Female	-0.018	-0.002	0.045
	(0.152)	(0.153)	(0.125)
Non-German	0.961***	0.283	1.031***
	(0.261)	(0.235)	(0.239)
Std. Errors	by name-address cluster	by name-address cluster	by name-address cluster
n	31,516	41,548	41,548

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S4 uses the same specification but estimates primary vaccinations (first and second dose) and booster vaccinations separately.

[4] This alternative is also imperfect as it does not take into account all variants of hyphenated names or co-residents with different last names.

Table S4: Main results with clustered standard errors defined as shared name and address (by vaccination type). Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations.

	Primary vaccination		**Booster vaccination**	
	(1) Spillover effect	(2) Overall effect	(3) Spillover effect	(4) Overall effect
Treatment	0.021	0.032	-0.295*	-0.299**
	(0.107)	(0.090)	(0.123)	(0.112)
Age	-0.026	-0.026+	0.041*	0.043**
	(0.016)	(0.014)	(0.017)	(0.016)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.228*	-0.184*	0.208+	0.231*
	(0.101)	(0.080)	(0.116)	(0.098)
Non-German	1.444***	1.460***	-0.478***	-0.420**
	(0.222)	(0.203)	(0.141)	(0.130)
Std. Errors	by name-address cluster	by name-address cluster	by name-address cluster	by name-address cluster
n	31,163	41,059	31,259	41,241

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Though we did not pre-register additional cluster specifications, for completeness we include two additional analyses. In the subsequent subsections, we present our main results with precinct-cluster standard errors and another analysis where we limit the sample to residents of small clusters only.

5.1.2 Precincts as Clusters

With this specification, we use the 60 precincts in the city to calculate clustered standard errors. Table S5 provides the main results with this alternative specification.

Table S5: Main results with precincts as clusters. Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations and information uptake.

	(1) Spillover effect Vaccinations	(2) Overall effect Clicks	(3) Overall effect Vaccinations
Treatment	-0.269	0.000	-0.263+
	(0.162)	(0.108)	(0.153)
Age	0.014	0.140***	0.017
	(0.014)	(0.009)	(0.012)
Age^2	0.000	-0.001***	0.000
	(0.000)	(0.000)	(0.000)
Female	-0.018	-0.002	0.045
	(0.096)	(0.099)	(0.076)
Non-German	0.961***	0.283+	1.031***
	(0.237)	(0.145)	(0.217)
Std. Errors	by precinct cluster	by precinct cluster	by precinct cluster
n	31,516	41,548	41,548

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S6 also uses the precincts as social networks but estimates primary vaccinations (first and second dose) and booster vaccinations separately.

Table S6: Main results with precincts as clusters (by vaccination type). Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations.

	Primary vaccination		**Booster vaccination**	
	(1) Spillover effect	(2) Overall effect	(3) Spillover effect	(4) Overall effect
Treatment	0.021	0.032	-0.295***	-0.299***
	(0.125)	(0.103)	(0.075)	(0.082)
Age	-0.026*	-0.026**	0.041***	0.043***
	(0.011)	(0.010)	(0.011)	(0.010)
Age^2	0.000	0.000	0.000*	0.000*
	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.228***	-0.184***	0.208*	0.231**
	(0.041)	(0.038)	(0.101)	(0.071)
Non-German	1.444***	1.460***	-0.478***	-0.420***
	(0.245)	(0.223)	(0.043)	(0.045)
Std. Errors	by precinct cluster	by precinct cluster	by precinct cluster	by precinct cluster
n	31,163	41,059	31,259	41,241

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

5.1.3 Small Clusters

With this alternative specification, we restricted the sample to addresses with less than five residents. Table S7 depicts the main results.

Table S7: Main results with small clusters (less than five members) only. Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations and information uptake.

	(1) Spillover effect Vaccinations	(2) Overall effect Clicks	(3) Overall effect Vaccinations
Treatment	-0.516+	-0.443*	-0.501*
	(0.266)	(0.220)	(0.213)
Age	0.035	0.132***	0.040
	(0.041)	(0.036)	(0.032)
Age^2	0.000	-0.001***	0.000
	(0.000)	(0.000)	(0.000)
Female	0.398	-0.028	0.330+
	(0.282)	(0.244)	(0.186)
Non-German	0.700	0.695	0.944+
	(0.598)	(0.523)	(0.502)
Std. Errors	by address-cluster	by address-cluster	by address-cluster
n	10,130	17,658	17,658

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S8 shows the results for primary vaccinations (first and second dose) and booster vaccinations separately.

Table S8: Main results with small clusters (less than five members) only (by vaccination type). Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations.

	Primary vaccination		**Booster vaccination**	
	(1)Spillover effect	(2)Overall effect	(3)Spillover effect	(4)Overall effect
Treatment	0.228	0.113	-0.738**	-0.612**
	(0.142)	(0.105)	(0.227)	(0.187)
Age	-0.041	-0.035+	0.077*	0.074**
	(0.026)	(0.019)	(0.032)	(0.026)
Age^2	0.000	0.000	0.000	0.000+
	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.064	-0.023	0.465*	0.363*
	(0.164)	(0.099)	(0.233)	(0.159)
Non-German	1.241*	1.441***	-0.564	-0.499+
	(0.483)	(0.417)	(0.358)	(0.290)
Std. Errors	by address-cluster	by address-cluster	by address-cluster	by address-cluster
n	9,996	17,409	10,075	17,574

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Though our analysis of small clusters reduces our sample size, we recover a negative spillover and overall effect with a smaller p-value than in our main analysis. This evidence supports the interpretation that social interactions between recipients of the treatment letter and the control led to fewer control-letter cohabitants of treatment-letter recipients seeking vaccination at the advertised public events. As mentioned in the main text, there is suggestive evidence that these individuals sought out other vaccination opportunities, such as with the additional events not mentioned in the letters.

5.2 Alternative methods to code vaccination

As mentioned in Section 2, the city organized eleven additional vaccination events during and after the study period. These additional events were not mentioned in the initial letter, however, starting on November 15th, 2021, residents who visited the website or called the information hotline could learn about these additional events. 714 residents attended and were vaccinated at one of these additional events. In our main analysis, vaccination at these events is coded as unvaccinated. To illustrate the robustness of our main analysis, we use all possible methods to code vaccination here.

The following tables replicate our main analyses including all individuals who were vaccinated at additional events (i.e., if an individual was vaccinated at an additional event their vaccination status is set =1 instead of zero, as in our main analysis.)

Table S9: Treatment effects with regular and additional vaccination events coded as successful vaccinations. Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations.

	(1) Direct effect	(2) Spillover effect	(3) Overall effect
Treatment	-0.014	-0.200	-0.155
	(0.356)	(0.232)	(0.203)
Age	0.073	0.001	0.011
	(0.061)	(0.033)	(0.028)
Age^2	0.000	0.000	0.000
	(0.001)	(0.000)	(0.000)
Female	0.137	0.163	0.167
	(0.417)	(0.209)	(0.169)
Non-German	2.065**	1.674***	1.696***
	(0.782)	(0.399)	(0.361)
Std. Errors	Hetero-skedasticity-robust	by address cluster	by address cluster
n	10,032	31,516	41,548

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S10: Treatment effects with regular and additional vaccination events coded as successful vaccinations (by vaccination type). Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations. Effects are estimated for primary vaccinations (first and second dose) and booster vaccinations separately.

	Primary vaccination			Booster vaccination		
	(1) Direct Effect	(2) Spillover Effect	(3) Overall Effect	(4) Direct Effect	(5) Spillover Effect	(6) Overall Effect
Treatment	0.256	-0.063	0.014	-0.267	-0.132	-0.164
	(0.184)	(0.153)	(0.127)	(0.314)	(0.178)	(0.160)
Age	-0.030	-0.029	-0.031+	0.105*	0.033	0.042+
	(0.034)	(0.019)	(0.017)	(0.052)	(0.027)	(0.024)
Age^2	0.000	0.000	0.000	-0.001	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.176	-0.177	-0.155	0.323	0.350*	0.333*
	(0.226)	(0.124)	(0.100)	(0.360)	(0.171)	(0.140)
Non-German	2.554***	2.426***	2.442***	-0.469	-0.716**	-0.709**
	(0.574)	(0.325)	(0.297)	(0.563)	(0.250)	(0.224)
Std. Errors	Hetero-skedasticity-robust	by address cluster	by address cluster	Hetero-skedasticity-robust	by address cluster	by address cluster
n	9,786	30,738	40,524	9,950	31,118	41,068

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

5.3 P-value robustness

An alternative method to estimate p-values for our main analysis would be to use randomization inference (as described in e.g., Gerber & Green, 2012). The results are shown in Table S11.

Table S11: Randomization inference p-values. Results of 1,000 iterations of randomly permuting the treatment assignment vector, while keeping the rest of our data frame fixed

Effect Type	**p-value**
Direct effect (information uptake)	0.190
Direct effect (vaccinations)	0.198
Spillover effect (vaccinations)	0.075
Overall effect (information uptake)	0.483
Overall effect (vaccinations)	0.061

5.4 Treatment Effects Over Time

As pre-registered, we correct for multiple comparisons using the Holm-Bonferroni correction.

5.4.1 Effects by Weekends

Table S12: Direct treatment effects for each vaccination event weekend. Covariate-adjusted OLS regression.

	(1) Weekend 1	(2) Weekend 2	(3) Weekend 3	(4) Weekend 4
Treatment	-0.064	-0.048	-0.007	-0.122
	(0.120)	(0.150)	(0.144)	(0.135)
Age	0.022	0.004	0.012	-0.003
	(0.020)	(0.027)	(0.026)	(0.024)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.212	-0.010	-0.178	0.084
	(0.138)	(0.172)	(0.179)	(0.143)
Non-German	0.121	0.056	-0.070	1.175**
	(0.192)	(0.295)	(0.330)	(0.359)
n	9,881	9,901	9,896	9,892

Estimates are expressed in terms of percentage points. Heteroskedasticity-robust standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S13: Direct treatment effects on primary vaccinations for each vaccination event weekend. Covariate-adjusted OLS regression.

	(1) Weekend 1	(2) Weekend 2	(3) Weekend 3	(4) Weekend 4
Treatment	-0.042	0.115	0.034	-0.052
	(0.050)	(0.085)	(0.076)	(0.069)
Age	-0.009	-0.001	-0.004	-0.014
	(0.009)	(0.016)	(0.015)	(0.012)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.007	-0.138	-0.043	-0.003
	(0.051)	(0.108)	(0.102)	(0.076)
Non-German	0.244	0.216	0.232	0.842*
	(0.154)	(0.223)	(0.253)	(0.284)
n	9,852	9,864	9,860	9,858

Estimates are expressed in terms of percentage points. Heteroskedasticity-robust standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S14: Direct treatment effects on booster vaccinations for each vaccination event weekend. Covariate-adjusted OLS regression.

	(1) Weekend 1	(2) Weekend 2	(3) Weekend 3	(4) Weekend 4
Treatment	-0.022	-0.164	-0.040	-0.084
	(0.110)	(0.123)	(0.122)	(0.119)
Age	0.031	0.005	0.016	0.010
	(0.018)	(0.022)	(0.021)	(0.020)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.205	0.127	-0.136	0.097
	(0.129)	(0.134)	(0.148)	(0.126)
Non-German	-0.124	-0.160	-0.305	0.366
	(0.115)	(0.193)	(0.213)	(0.225)
n	9,875	9,883	9,882	9,880

Estimates are expressed in terms of percentage points. Heteroskedasticity-robust standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S15: Spillover treatment effects for each vaccination event weekend. Covariate-adjusted OLS regression.

	(1) Weekend 1	(2) Weekend 2	(3) Weekend 3	(4) Weekend 4
Treatment	-0.287*	-0.090	-0.045	0.139
	(0.102)	(0.077)	(0.083)	(0.082)
Age	0.013	-0.006	0.020	-0.013
	(0.010)	(0.012)	(0.013)	(0.013)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.012	-0.007	-0.044	0.015
	(0.064)	(0.076)	(0.089)	(0.078)
Non-German	0.136	0.216	0.246	0.405*
	(0.191)	(0.143)	(0.150)	(0.146)
Std. Errors	by address cluster	by address cluster	by address cluster	by address cluster
n	31,010	31,058	31,099	31,067

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S16: Spillover treatment effects on primary vaccinations for each vaccination event weekend. Covariate-adjusted OLS regression.

	(1) Weekend 1	(2) Weekend 2	(3) Weekend 3	(4) Weekend 4
Treatment	-0.116	0.049	0.029	0.060
	(0.093)	(0.049)	(0.057)	(0.047)
Age	-0.003	-0.013	0.008	-0.019+
	(0.006)	(0.009)	(0.009)	(0.008)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.053	-0.022	-0.102	-0.056
	(0.045)	(0.049)	(0.060)	(0.048)
Non-German	0.244	0.365**	0.369**	0.510***
	(0.188)	(0.122)	(0.126)	(0.120)
n	30,956	30,969	30,992	30,964

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S17: Spillover treatment effects on booster vaccinations for each vaccination event weekend. Covariate-adjusted OLS regression.

	(1) Weekend 1	(2) Weekend 2	(3) Weekend 3	(4) Weekend 4
Treatment	-0.172***	-0.141+	-0.075	0.080
	(0.044)	(0.060)	(0.062)	(0.066)
Age	0.015	0.007	0.012	0.007
	(0.008)	(0.009)	(0.009)	(0.010)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.065	0.015	0.057	0.070
	(0.046)	(0.058)	(0.066)	(0.061)
Non-German	-0.107*	-0.146	-0.122	-0.104
	(0.036)	(0.075)	(0.083)	(0.082)
n	30,960	30,995	31,013	31,009

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S18: Overall treatment effects for each vaccination event weekend. Covariate-adjusted OLS regression.

	(1) Weekend 1	(2) Weekend 2	(3) Weekend 3	(4) Weekend 4
Treatment	-0.232*	-0.083	-0.037	0.078
	(0.086)	(0.071)	(0.074)	(0.072)
Age	0.014	-0.004	0.016	-0.009
	(0.009)	(0.011)	(0.011)	(0.011)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.060	0.025	-0.066	0.025
	(0.056)	(0.064)	(0.071)	(0.063)
Non-German	0.124	0.200	0.222	0.524***
	(0.173)	(0.127)	(0.135)	(0.141)
Std. Errors	by address cluster	by address cluster	by address cluster	by address cluster
n	40,891	40,959	40,995	40,959

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S19: Overall treatment effects on primary vaccinations for each vaccination event weekend. Covariate-adjusted OLS regression.

	(1) Weekend 1	(2) Weekend 2	(3) Weekend 3	(4) Weekend 4
Treatment	-0.097	0.062	0.031	0.036
	(0.075)	(0.043)	(0.047)	(0.041)
Age	-0.004	-0.010	0.004	-0.017+
	(0.005)	(0.007)	(0.008)	(0.007)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.033	-0.035	-0.077	-0.044
	(0.036)	(0.041)	(0.047)	(0.037)
Non-German	0.239	0.345**	0.360**	0.558***
	(0.169)	(0.108)	(0.113)	(0.116)
n	40,808	40,833	40,852	40,822

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S20: Overall treatment effects on booster vaccinations for each vaccination event weekend. Covariate-adjusted OLS regression.

	(1) Weekend 1	(2) Weekend 2	(3) Weekend 3	(4) Weekend 4
Treatment	-0.135**	-0.147*	-0.069	0.042
	(0.042)	(0.056)	(0.058)	(0.059)
Age	0.019*	0.007	0.011	0.007
	(0.007)	(0.008)	(0.008)	(0.009)
Age^2	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.092	0.061	0.011	0.069
	(0.043)	(0.050)	(0.053)	(0.050)
Non-German	-0.115**	-0.143+	-0.138	-0.030
	(0.034)	(0.067)	(0.074)	(0.080)
n	40,835	40,878	40,895	40,889

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

5.4.2 Effects by Day

Table S21: Direct treatment effects for each vaccination event. Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.064	-0.033	-0.015	0.020	-0.027	0.004	-0.139
	(0.120)	(0.103)	(0.109)	(0.114)	(0.088)	(0.099)	(0.096)
Age	0.022	0.011	-0.007	0.013	-0.001	0.001	-0.005
	(0.020)	(0.016)	(0.022)	(0.019)	(0.017)	(0.017)	(0.017)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.212	-0.269	0.259	-0.123	-0.056	0.025	0.069
	(0.138)	(0.110)	(0.132)	(0.148)	(0.101)	(0.108)	(0.100)
Non-German	0.121	-0.087	0.144	-0.214	0.142	0.540	0.670
	(0.192)	(0.224)	(0.192)	(0.232)	(0.237)	(0.234)	(0.278)
n	9,881	9,872	9,875	9,877	9,865	9,870	9,868

Estimates are expressed in terms of percentage points. Heteroskedasticity-robust standard errors in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S22: Direct treatment effects on primary vaccinations for each vaccination event. Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.042	0.082	0.033	0.041	-0.007	0.005	-0.057
	(0.050)	(0.058)	(0.063)	(0.041)	(0.064)	(0.038)	(0.057)
Age	-0.009	0.002	-0.003	0.004	-0.008	-0.011	-0.003
	(0.009)	(0.008)	(0.014)	(0.005)	(0.014)	(0.008)	(0.009)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.007	-0.162	0.024	-0.028	-0.015	-0.071	0.068
	(0.051)	(0.072)	(0.080)	(0.057)	(0.084)	(0.042)	(0.063)
Non-German	0.244	0.014	0.203	0.001	0.231	0.359	0.487
	(0.154)	(0.137)	(0.177)	(0.119)	(0.224)	(0.178)	(0.224)
n	9,852	9,854	9,856	9,850	9,856	9,850	9,854

Estimates are expressed in terms of percentage points. Heteroskedasticity-robust standard errors in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S23: Direct treatment effects on booster vaccinations for each vaccination event. Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.022	-0.116	-0.048	-0.020	-0.020	-0.002	-0.083
	(0.110)	(0.086)	(0.089)	(0.106)	(0.061)	(0.091)	(0.077)
Age	0.031	0.009	-0.004	0.009	0.007	0.012	-0.002
	(0.018)	(0.014)	(0.017)	(0.019)	(0.009)	(0.015)	(0.014)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.205	-0.108	0.235	-0.095	-0.042	0.096	0.001
	(0.129)	(0.083)	(0.105)	(0.137)	(0.057)	(0.099)	(0.077)
Non-German	-0.124	-0.102	-0.059	-0.216	-0.090	0.182	0.184
	(0.115)	(0.178)	(0.075)	(0.200)	(0.075)	(0.152)	(0.167)
n	9,875	9,864	9,865	9,873	9,855	9,866	9,860

Estimates are expressed in terms of percentage points. Heteroskedasticity-robust standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S24: Spillover treatment effects for each vaccination event. Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.064	-0.033	-0.015	0.020	-0.027	0.004	-0.139
	(0.120)	(0.103)	(0.109)	(0.114)	(0.088)	(0.099)	(0.096)
Age	0.022	0.011	-0.007	0.013	-0.001	0.001	-0.005
	(0.020)	(0.016)	(0.022)	(0.019)	(0.017)	(0.017)	(0.017)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.212	-0.269	0.259	-0.123	-0.056	0.025	0.069
	(0.138)	(0.110)	(0.132)	(0.148)	(0.101)	(0.108)	(0.100)
Non-German	0.121	-0.087	0.144	-0.214	0.142	0.540	0.670
	(0.192)	(0.224)	(0.192)	(0.232)	(0.237)	(0.234)	(0.278)
n	9,881	9,872	9,875	9,877	9,865	9,870	9,868

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S25: Spillover treatment effects on primary vaccinations for each vaccination event. Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.042	0.082	0.033	0.041	-0.007	0.005	-0.057
	(0.050)	(0.058)	(0.063)	(0.041)	(0.064)	(0.038)	(0.057)
Age	-0.009	0.002	-0.003	0.004	-0.008	-0.011	-0.003
	(0.009)	(0.008)	(0.014)	(0.005)	(0.014)	(0.008)	(0.009)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.007	-0.162	0.024	-0.028	-0.015	-0.071	0.068
	(0.051)	(0.072)	(0.080)	(0.057)	(0.084)	(0.042)	(0.063)
Non-German	0.244	0.014	0.203	0.001	0.231	0.359	0.487
	(0.154)	(0.137)	(0.177)	(0.119)	(0.224)	(0.178)	(0.224)
n	9,852	9,854	9,856	9,850	9,856	9,850	9,854

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S26: Spillover treatment effects on booster vaccinations for each vaccination event. Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.022	-0.116	-0.048	-0.020	-0.020	-0.002	-0.083
	(0.110)	(0.086)	(0.089)	(0.106)	(0.061)	(0.091)	(0.077)
Age	0.031	0.009	-0.004	0.009	0.007	0.012	-0.002
	(0.018)	(0.014)	(0.017)	(0.019)	(0.009)	(0.015)	(0.014)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.205	-0.108	0.235	-0.095	-0.042	0.096	0.001
	(0.129)	(0.083)	(0.105)	(0.137)	(0.057)	(0.099)	(0.077)
Non-German	-0.124	-0.102	-0.059	-0.216	-0.090	0.182	0.184
	(0.115)	(0.178)	(0.075)	(0.200)	(0.075)	(0.152)	(0.167)
n	9,875	9,864	9,865	9,873	9,855	9,866	9,860

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S27: Overall treatment effects for each vaccination event. Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.064	-0.033	-0.015	0.020	-0.027	0.004	-0.139
	(0.120)	(0.103)	(0.109)	(0.114)	(0.088)	(0.099)	(0.096)
Age	0.022	0.011	-0.007	0.013	-0.001	0.001	-0.005
	(0.020)	(0.016)	(0.022)	(0.019)	(0.017)	(0.017)	(0.017)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.212	-0.269	0.259	-0.123	-0.056	0.025	0.069
	(0.138)	(0.110)	(0.132)	(0.148)	(0.101)	(0.108)	(0.100)
Non-German	0.121	-0.087	0.144	-0.214	0.142	0.540	0.670
	(0.192)	(0.224)	(0.192)	(0.232)	(0.237)	(0.234)	(0.278)
n	9,881	9,872	9,875	9,877	9,865	9,870	9,868

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S28: Overall treatment effects on primary vaccinations for each vaccination event. Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.042	0.082	0.033	0.041	-0.007	0.005	-0.057
	(0.050)	(0.058)	(0.063)	(0.041)	(0.064)	(0.038)	(0.057)
Age	-0.009	0.002	-0.003	0.004	-0.008	-0.011	-0.003
	(0.009)	(0.008)	(0.014)	(0.005)	(0.014)	(0.008)	(0.009)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.007	-0.162	0.024	-0.028	-0.015	-0.071	0.068
	(0.051)	(0.072)	(0.080)	(0.057)	(0.084)	(0.042)	(0.063)
Non-German	0.244	0.014	0.203	0.001	0.231	0.359	0.487
	(0.154)	(0.137)	(0.177)	(0.119)	(0.224)	(0.178)	(0.224)
n	9,852	9,854	9,856	9,850	9,856	9,850	9,854

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S29: Overall treatment effects on booster vaccinations for each vaccination event. Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.022	-0.116	-0.048	-0.020	-0.020	-0.002	-0.083
	(0.110)	(0.086)	(0.089)	(0.106)	(0.061)	(0.091)	(0.077)
Age	0.031	0.009	-0.004	0.009	0.007	0.012	-0.002
	(0.018)	(0.014)	(0.017)	(0.019)	(0.009)	(0.015)	(0.014)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.205	-0.108	0.235	-0.095	-0.042	0.096	0.001
	(0.129)	(0.083)	(0.105)	(0.137)	(0.057)	(0.099)	(0.077)
Non-German	-0.124	-0.102	-0.059	-0.216	-0.090	0.182	0.184
	(0.115)	(0.178)	(0.075)	(0.200)	(0.075)	(0.152)	(0.167)
n	9,875	9,864	9,865	9,873	9,855	9,866	9,860

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

5.4.3 Effects by Day (Cumulative)

Table S30: Direct treatment effects for each vaccination event (Cumulative). Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.064	-0.096	-0.124	-0.103	-0.130	-0.126	-0.247
	(0.120)	(0.158)	(0.190)	(0.220)	(0.236)	(0.254)	(0.269)
Age	0.022	0.033	0.025	0.038	0.037	0.037	0.033
	(0.020)	(0.026)	(0.034)	(0.039)	(0.042)	(0.045)	(0.048)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.212	-0.057	0.192	0.069	0.014	0.054	0.112
	(0.138)	(0.176)	(0.219)	(0.262)	(0.280)	(0.297)	(0.310)
Non-German	0.121	0.033	0.174	-0.038	0.105	0.611	1.235
	(0.192)	(0.295)	(0.350)	(0.418)	(0.477)	(0.526)	(0.587)
n	9,881	9,907	9,936	9,967	9,986	10,010	10,032

Estimates are expressed in terms of percentage points. Heteroskedasticity-robust standard errors in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S31: Direct treatment effects on primary vaccinations for each vaccination event (Cumulative). Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.042	0.041	0.074	0.114	0.107	0.113	0.056
	(0.050)	(0.076)	(0.099)	(0.107)	(0.124)	(0.130)	(0.142)
Age	-0.009	-0.007	-0.010	-0.006	-0.013	-0.025	-0.028
	(0.009)	(0.012)	(0.018)	(0.019)	(0.023)	(0.024)	(0.026)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.007	-0.155	-0.131	-0.159	-0.173	-0.226	-0.159
	(0.051)	(0.088)	(0.119)	(0.132)	(0.156)	(0.159)	(0.171)
Non-German	0.244	0.257	0.458	0.459	0.685	1.010+	1.483**
	(0.154)	(0.206)	(0.270)	(0.295)	(0.368)	(0.404)	(0.457)
n	9,852	9,860	9,870	9,874	9,884	9,888	9,896

Estimates are expressed in terms of percentage points. Heteroskedasticity-robust standard errors in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S32: Direct treatment effects on booster vaccinations for each vaccination event (Cumulative). Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.022	-0.137	-0.198	-0.217	-0.237	-0.237	-0.319
	(0.110)	(0.139)	(0.163)	(0.193)	(0.202)	(0.221)	(0.233)
Age	0.031	0.040	0.035	0.044	0.051	0.063	0.061
	(0.018)	(0.023)	(0.029)	(0.034)	(0.035)	(0.038)	(0.040)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.205	0.098	0.322	0.227	0.185	0.279	0.279
	(0.129)	(0.153)	(0.185)	(0.228)	(0.235)	(0.254)	(0.264)
Non-German	-0.124	-0.225	-0.286	-0.501	-0.589	-0.407	-0.224
	(0.115)	(0.212)	(0.225)	(0.299)	(0.308)	(0.343)	(0.380)
n	9,875	9,893	9,912	9,939	9,948	9,968	9,982

Estimates are expressed in terms of percentage points. Heteroskedasticity-robust standard errors in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S33: Spillover treatment effects for each vaccination event (Cumulative). Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.064	-0.096	-0.124	-0.103	-0.130	-0.126	-0.247
	(0.120)	(0.158)	(0.190)	(0.220)	(0.236)	(0.254)	(0.269)
Age	0.022	0.033	0.025	0.038	0.037	0.037	0.033
	(0.020)	(0.026)	(0.034)	(0.039)	(0.042)	(0.045)	(0.048)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.212	-0.057	0.192	0.069	0.014	0.054	0.112
	(0.138)	(0.176)	(0.219)	(0.262)	(0.280)	(0.297)	(0.310)
Non-German	0.121	0.033	0.174	-0.038	0.105	0.611	1.235
	(0.192)	(0.295)	(0.350)	(0.418)	(0.477)	(0.526)	(0.587)
n	9,881	9,907	9,936	9,967	9,986	10,010	10,032

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S34: Spillover treatment effects on primary vaccinations for each vaccination event (Cumulative). Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.042	0.041	0.074	0.114	0.107	0.113	0.056
	(0.050)	(0.076)	(0.099)	(0.107)	(0.124)	(0.130)	(0.142)
Age	-0.009	-0.007	-0.010	-0.006	-0.013	-0.025	-0.028
	(0.009)	(0.012)	(0.018)	(0.019)	(0.023)	(0.024)	(0.026)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.007	-0.155	-0.131	-0.159	-0.173	-0.226	-0.159
	(0.051)	(0.088)	(0.119)	(0.132)	(0.156)	(0.159)	(0.171)
Non-German	0.244	0.257	0.458	0.459	0.685	1.010+	1.483**
	(0.154)	(0.206)	(0.270)	(0.295)	(0.368)	(0.404)	(0.457)
n	9,852	9,860	9,870	9,874	9,884	9,888	9,896

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S35: Spillover treatment effects on booster vaccinations for each vaccination event (Cumulative). Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.022	-0.137	-0.198	-0.217	-0.237	-0.237	-0.319
	(0.110)	(0.139)	(0.163)	(0.193)	(0.202)	(0.221)	(0.233)
Age	0.031	0.040	0.035	0.044	0.051	0.063	0.061
	(0.018)	(0.023)	(0.029)	(0.034)	(0.035)	(0.038)	(0.040)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.205	0.098	0.322	0.227	0.185	0.279	0.279
	(0.129)	(0.153)	(0.185)	(0.228)	(0.235)	(0.254)	(0.264)
Non-German	-0.124	-0.225	-0.286	-0.501	-0.589	-0.407	-0.224
	(0.115)	(0.212)	(0.225)	(0.299)	(0.308)	(0.343)	(0.380)
n	9,875	9,893	9,912	9,939	9,948	9,968	9,982

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S36: Overall treatment effects for each vaccination event (Cumulative). Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.064	-0.096	-0.124	-0.103	-0.130	-0.126	-0.247
	(0.120)	(0.158)	(0.190)	(0.220)	(0.236)	(0.254)	(0.269)
Age	0.022	0.033	0.025	0.038	0.037	0.037	0.033
	(0.020)	(0.026)	(0.034)	(0.039)	(0.042)	(0.045)	(0.048)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.212	-0.057	0.192	0.069	0.014	0.054	0.112
	(0.138)	(0.176)	(0.219)	(0.262)	(0.280)	(0.297)	(0.310)
Non-German	0.121	0.033	0.174	-0.038	0.105	0.611	1.235
	(0.192)	(0.295)	(0.350)	(0.418)	(0.477)	(0.526)	(0.587)
n	9,881	9,907	9,936	9,967	9,986	10,010	10,032

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S37: Overall treatment effects on primary vaccinations for each vaccination event (Cumulative). Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.042	0.041	0.074	0.114	0.107	0.113	0.056
	(0.050)	(0.076)	(0.099)	(0.107)	(0.124)	(0.130)	(0.142)
Age	-0.009	-0.007	-0.010	-0.006	-0.013	-0.025	-0.028
	(0.009)	(0.012)	(0.018)	(0.019)	(0.023)	(0.024)	(0.026)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.007	-0.155	-0.131	-0.159	-0.173	-0.226	-0.159
	(0.051)	(0.088)	(0.119)	(0.132)	(0.156)	(0.159)	(0.171)
Non-German	0.244	0.257	0.458	0.459	0.685	1.010+	1.483**
	(0.154)	(0.206)	(0.270)	(0.295)	(0.368)	(0.404)	(0.457)
n	9,852	9,860	9,870	9,874	9,884	9,888	9,896

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table S38: Overall treatment effects on booster vaccinations for each vaccination event (Cumulative). Covariate-adjusted OLS regression.

	(1) Event 1	(2) Event 2	(3) Event 3	(4) Event 4	(5) Event 5	(6) Event 6	(7) Event 7
Treatment	-0.022	-0.137	-0.198	-0.217	-0.237	-0.237	-0.319
	(0.110)	(0.139)	(0.163)	(0.193)	(0.202)	(0.221)	(0.233)
Age	0.031	0.040	0.035	0.044	0.051	0.063	0.061
	(0.018)	(0.023)	(0.029)	(0.034)	(0.035)	(0.038)	(0.040)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	0.205	0.098	0.322	0.227	0.185	0.279	0.279
	(0.129)	(0.153)	(0.185)	(0.228)	(0.235)	(0.254)	(0.264)
Non-German	-0.124	-0.225	-0.286	-0.501	-0.589	-0.407	-0.224
	(0.115)	(0.212)	(0.225)	(0.299)	(0.308)	(0.343)	(0.380)
n	9,875	9,893	9,912	9,939	9,948	9,968	9,982

Estimates are expressed in terms of percentage points. Standard errors (clustered by address clusters) in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

6 Section 6: Statistical Power and Sensitivity Analysis

We estimated power as part of our pre-registration plan (see below, Section 9 Pre-registered Analysis Plan, Sample Size). For more realistic *post hoc* estimates of power, we used the observed Intraclass Correlation Coefficient (ICC) in the control group and the sample sizes in our data in order to calculate the minimum detectable effect (MDE) for each of our estimators (using the `power` command in Stata 15.1). The results are presented in Table S39.

Table S39: Probability of observing the treatment effect.

Effect Type	**MDE**	**Power**
Direct effect (information uptake)	0.010	80%
Direct effect (vaccinations)	0.011	83%
Spillover effect (information uptake)	0.005	79%
Spillover effect (vaccinations)	0.006	82%
Overall effect (information uptake)	0.005	80%
Overall effect (vaccinations)	0.006	83%

As recommended in (Lakens 2022), we also include sensitivity power analyses for all the effect types. As seen in Figures S3-S12, our study is very-well powered to detect even very small effects. While we use Cohen's d to illustrate effect sizes in a more standardized fashion, we caution against using default interpretations of Cohen's d from outside of a public policy context. While a d of 0.1 is often considered trivial in the fields of psychology and education (e.g., Lakens, Scheel, Isager 2018), a d of 0.1 can be highly consequential in a public health context (e.g., Lakens 2013). The effect sizes we observed in our data are depicted in red and we note the effect size and power for all statistically significant effects at p=0.1.

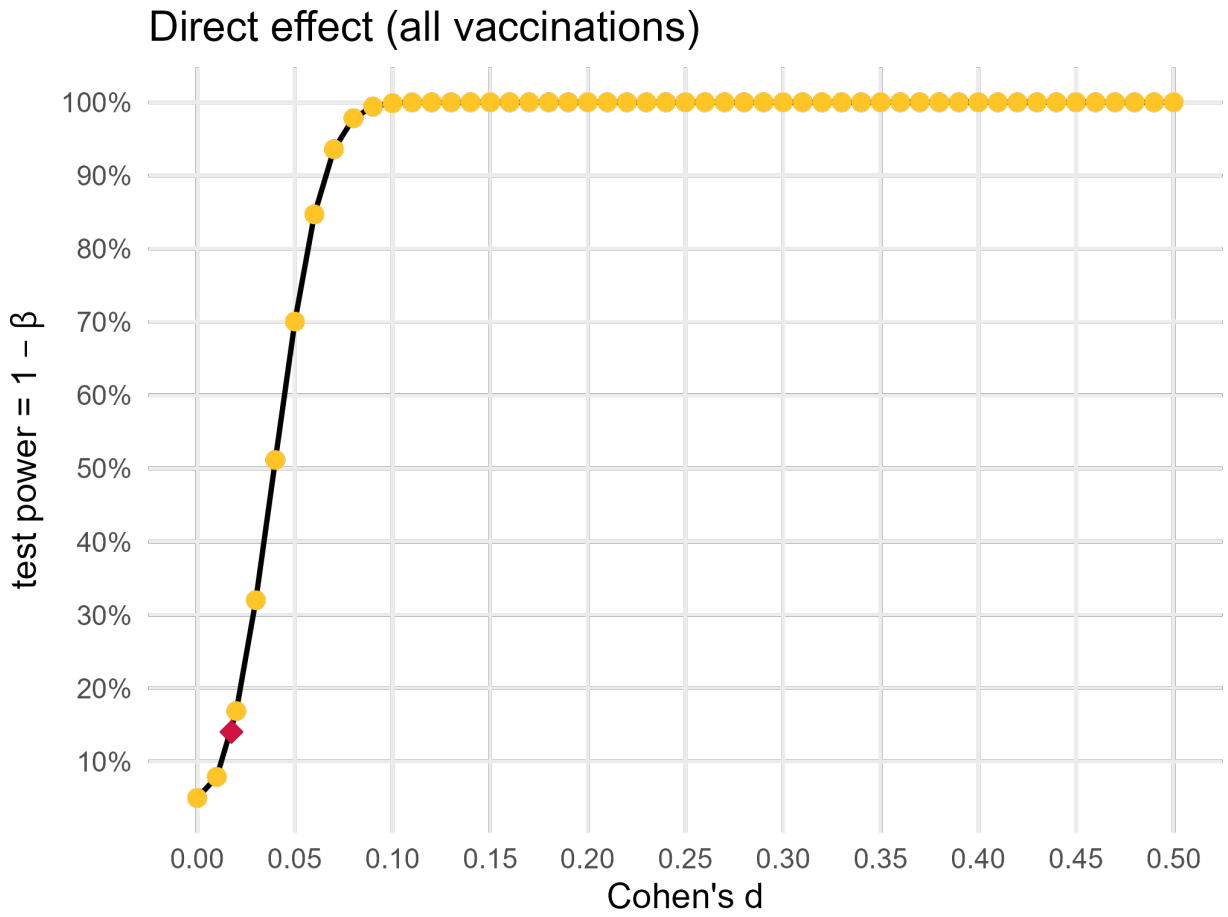


Figure S3: Sensitivity analysis of the direct effect on all vaccinations.

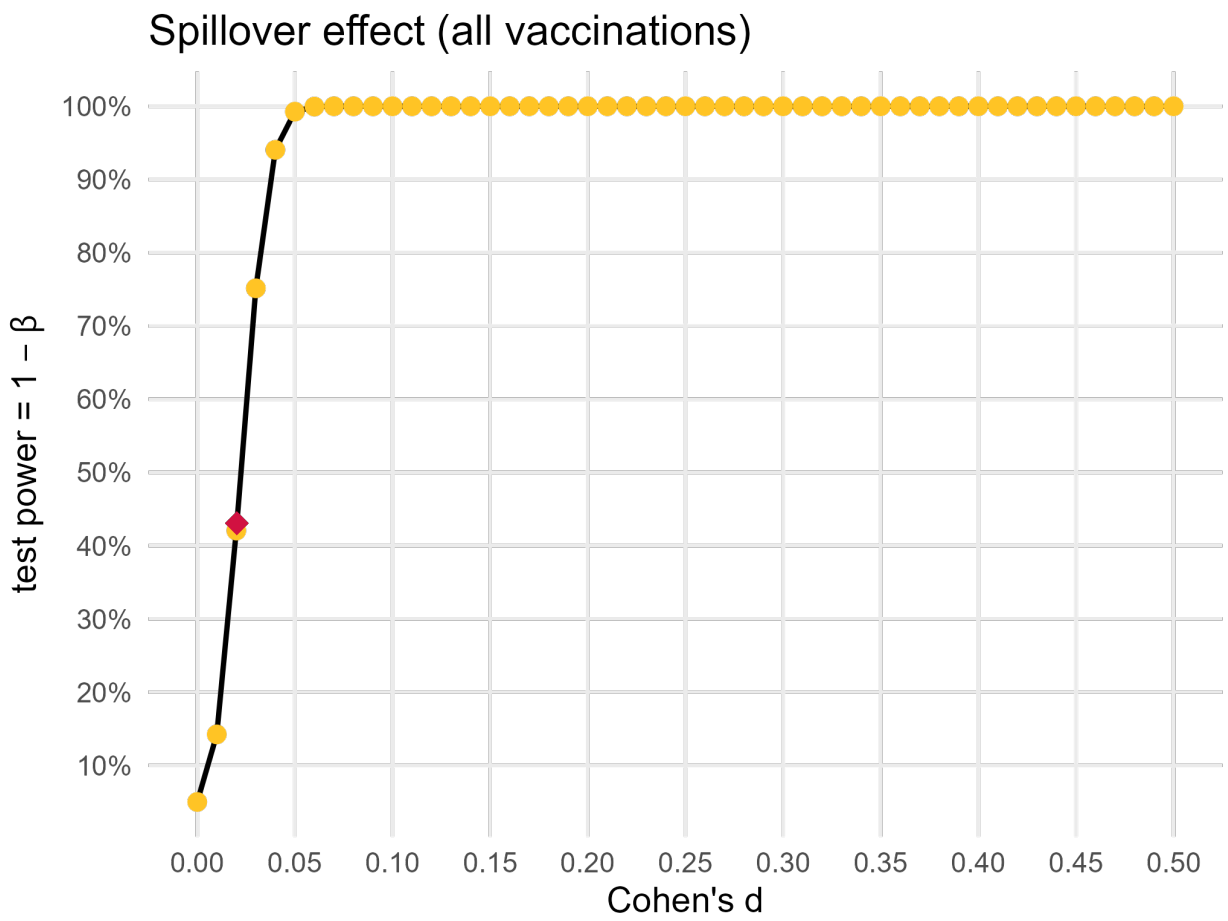


Figure S4: Sensitivity analysis of the spillover effect on all vaccinations.

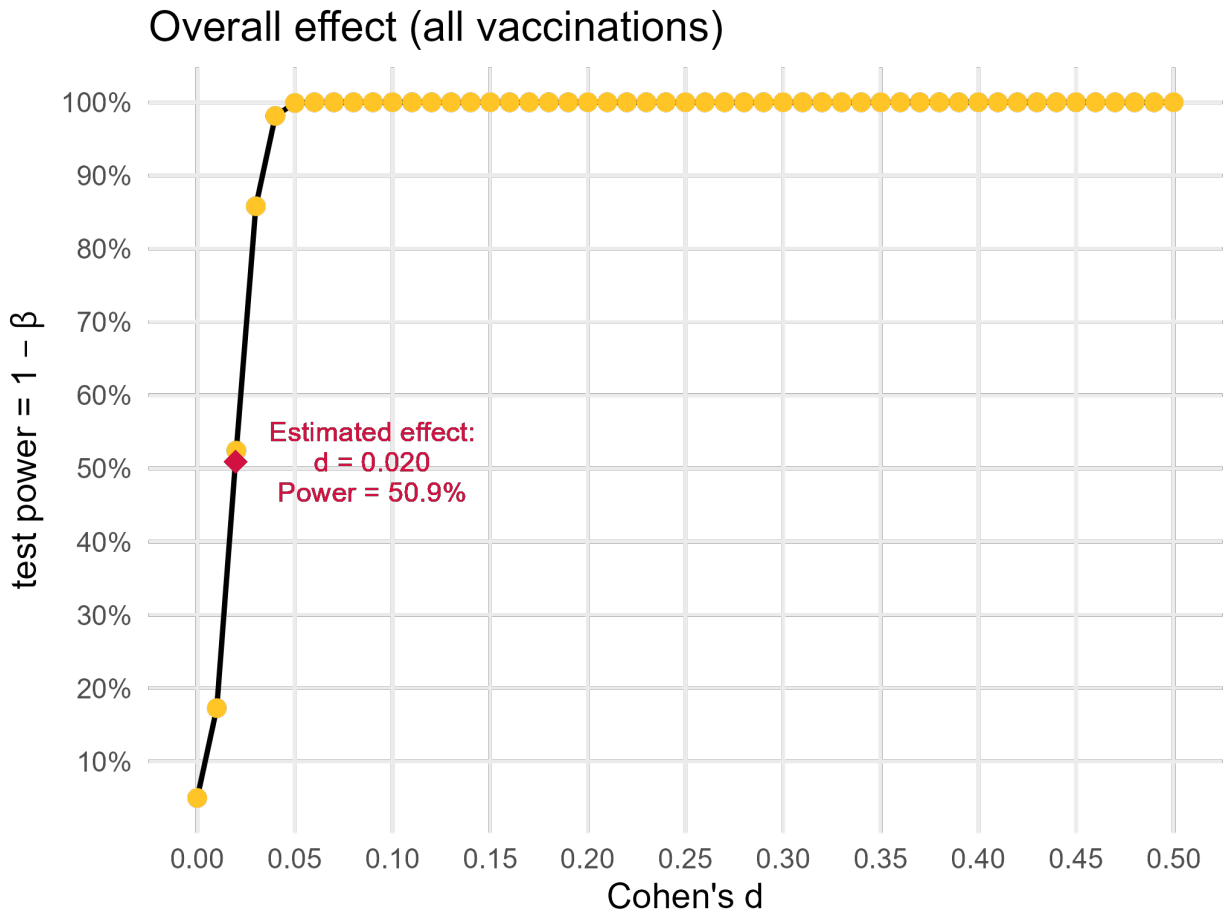


Figure S5: Sensitivity analysis of the overall effect on all vaccinations.

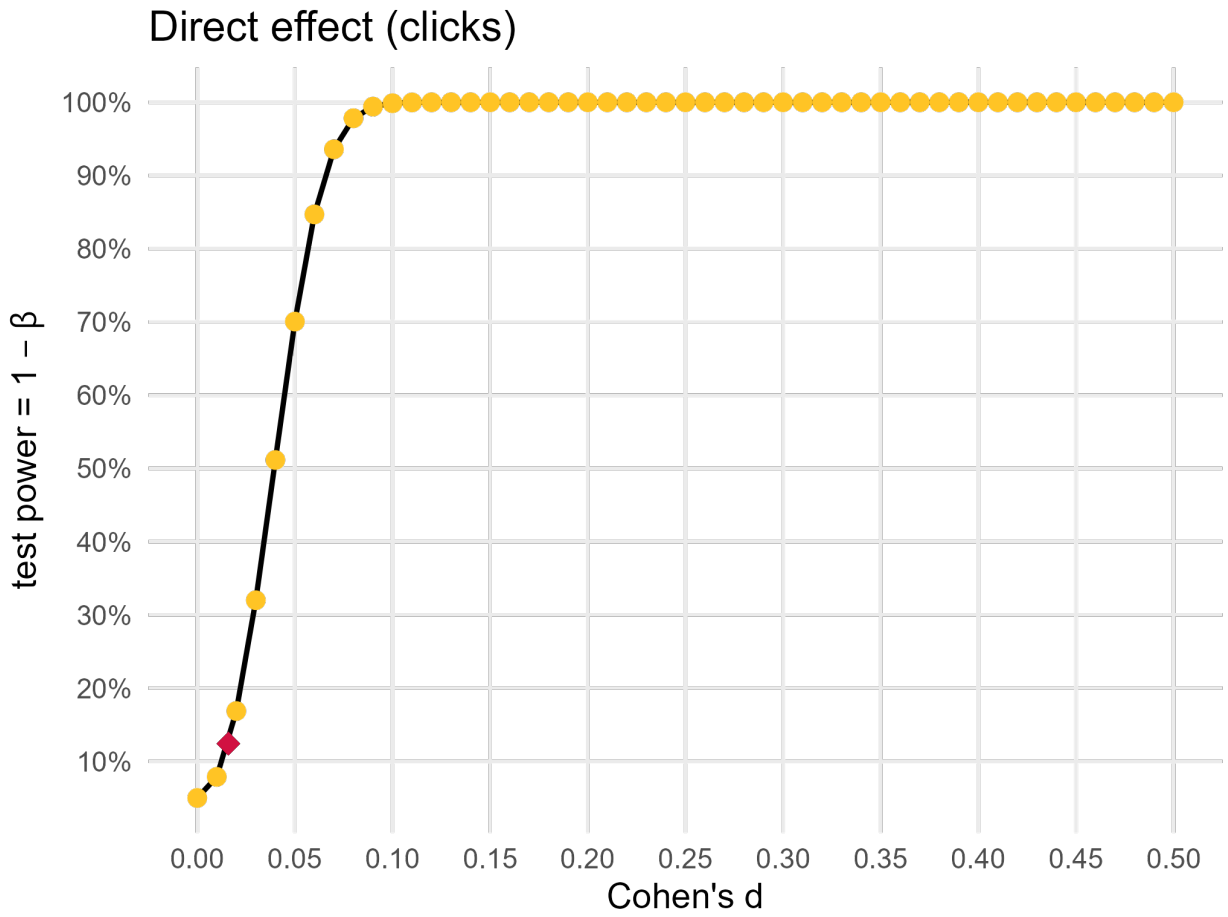


Figure S6: Sensitivity analysis of the direct effect on all information uptake (clicks).

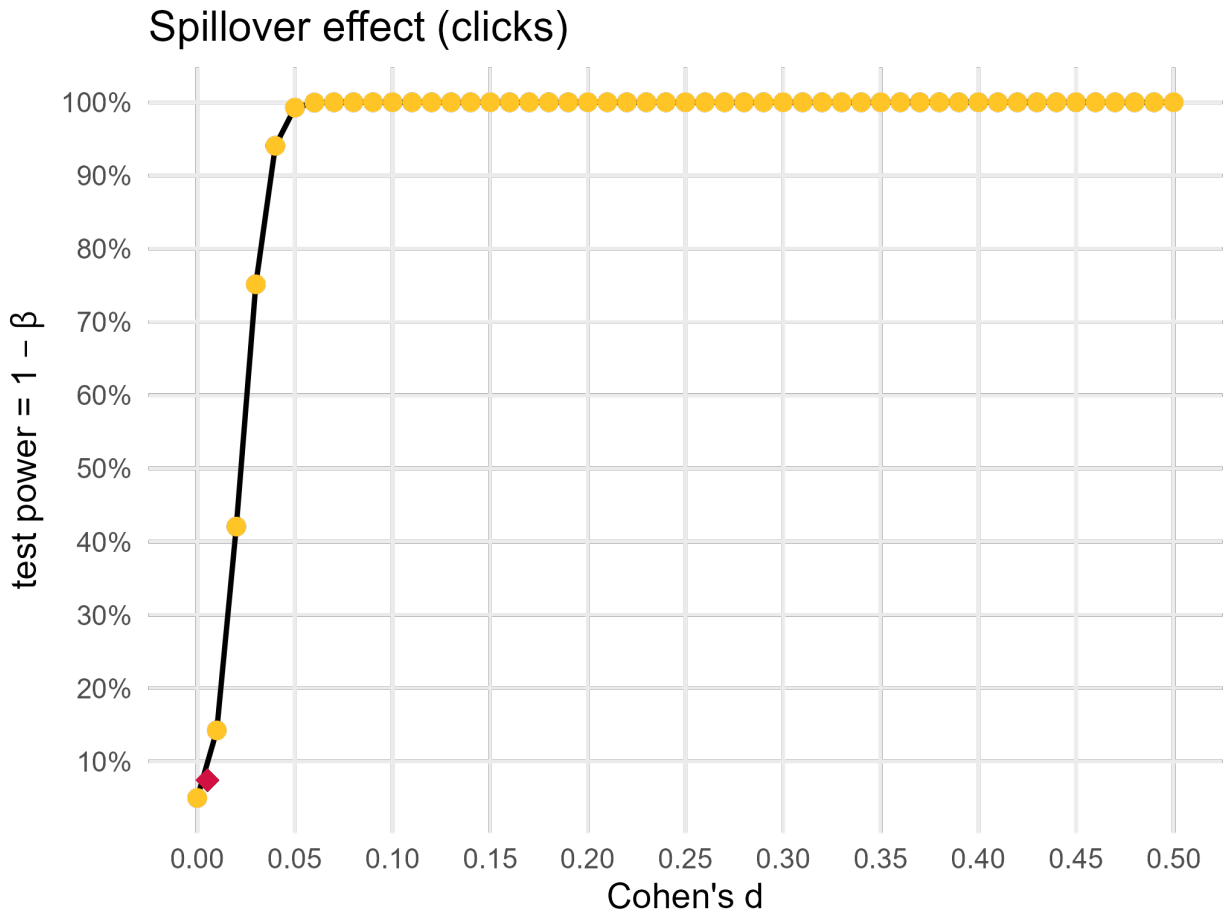


Figure S7: Sensitivity analysis of the spillover effect on all information uptake (clicks).

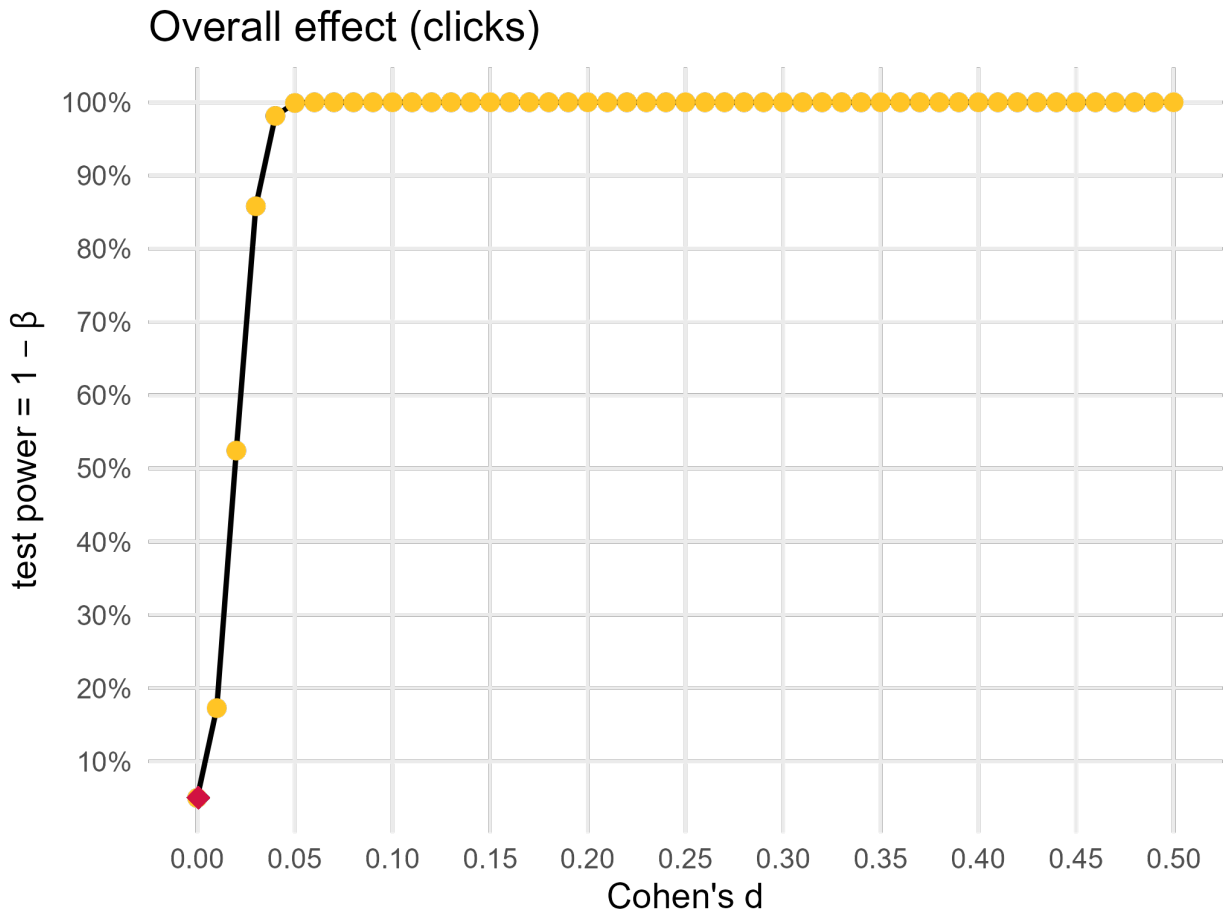


Figure S8: Sensitivity analysis of the overall effect on information uptake (clicks).

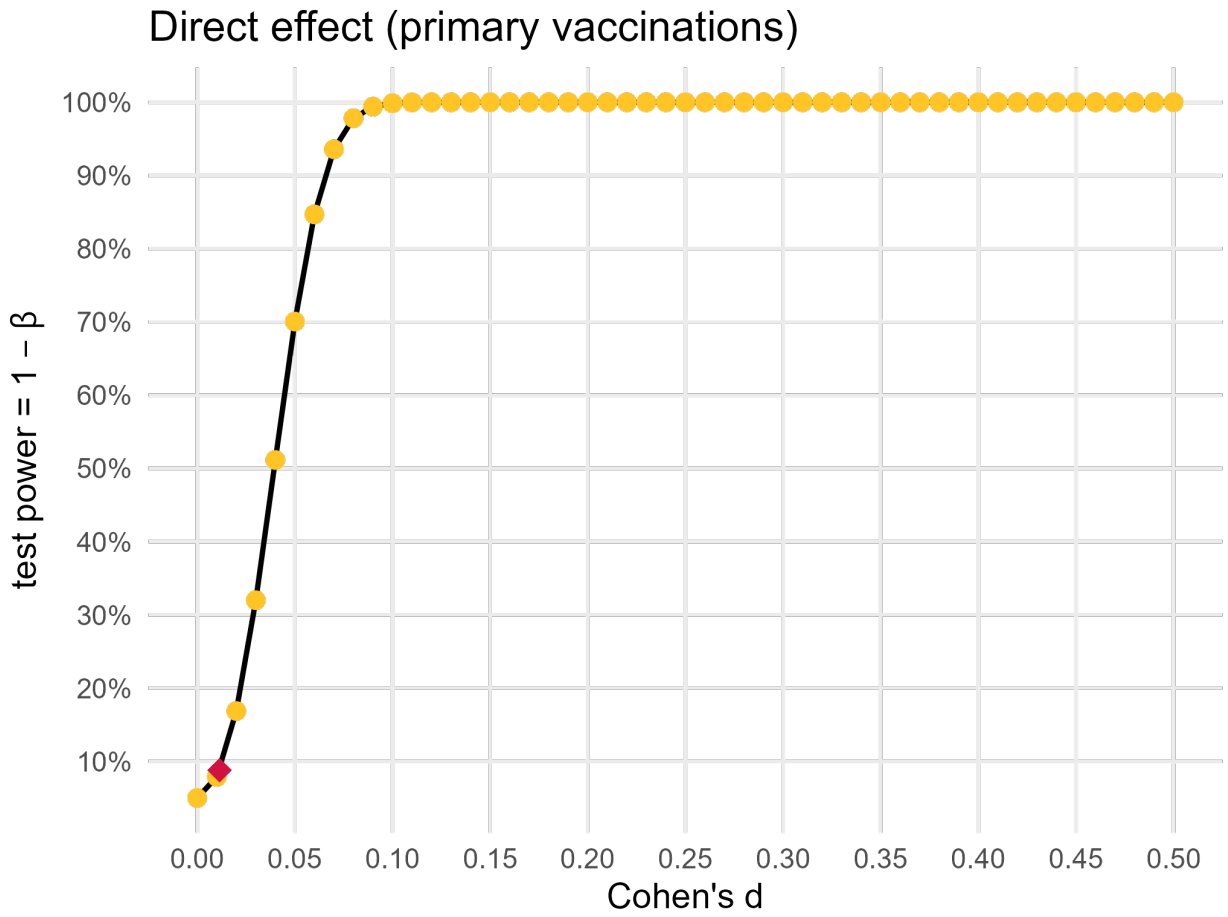


Figure S9: Sensitivity analysis of the direct effect on primary vaccinations.

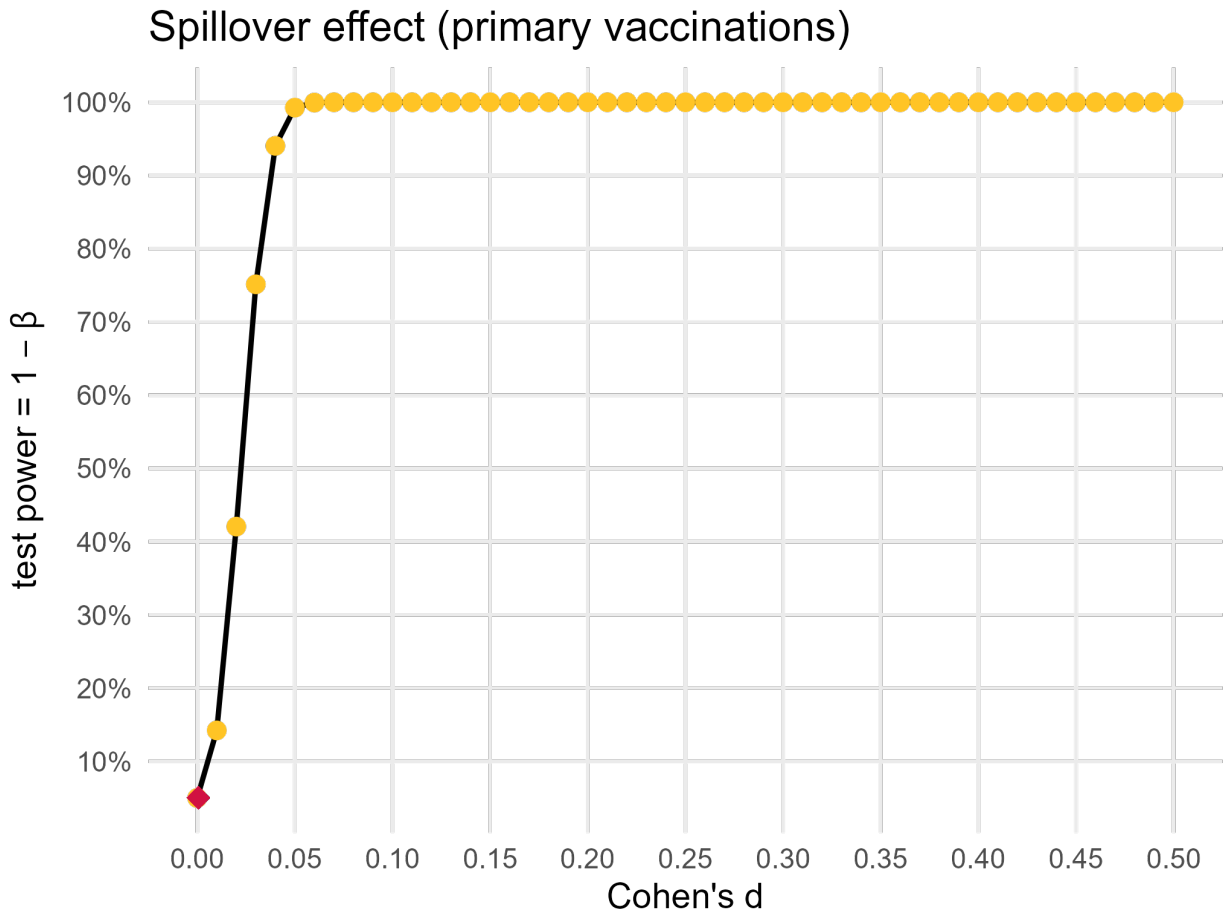


Figure S10: Sensitivity analysis of the spillover effect on primary vaccinations.

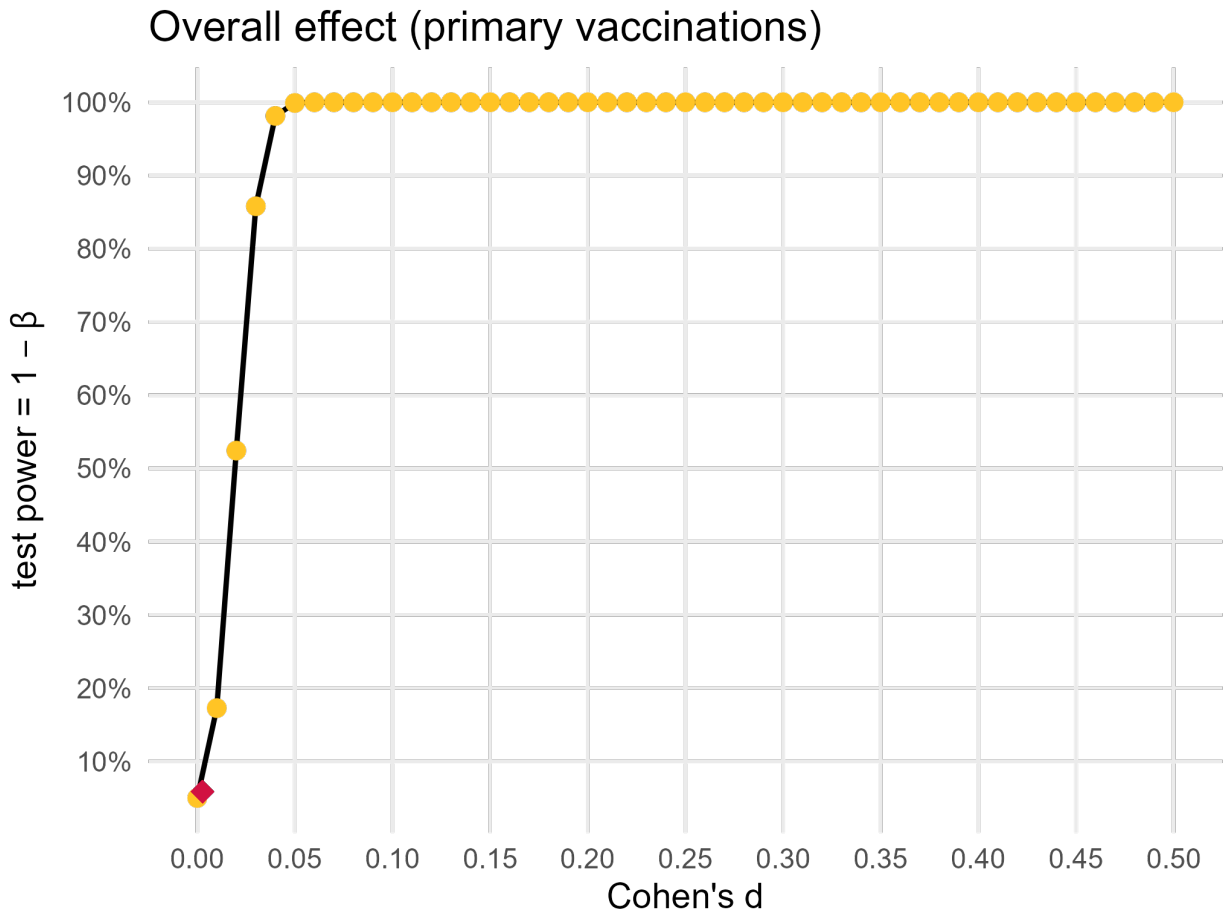


Figure S11: Sensitivity analysis of the overall effect on primary vaccinations.

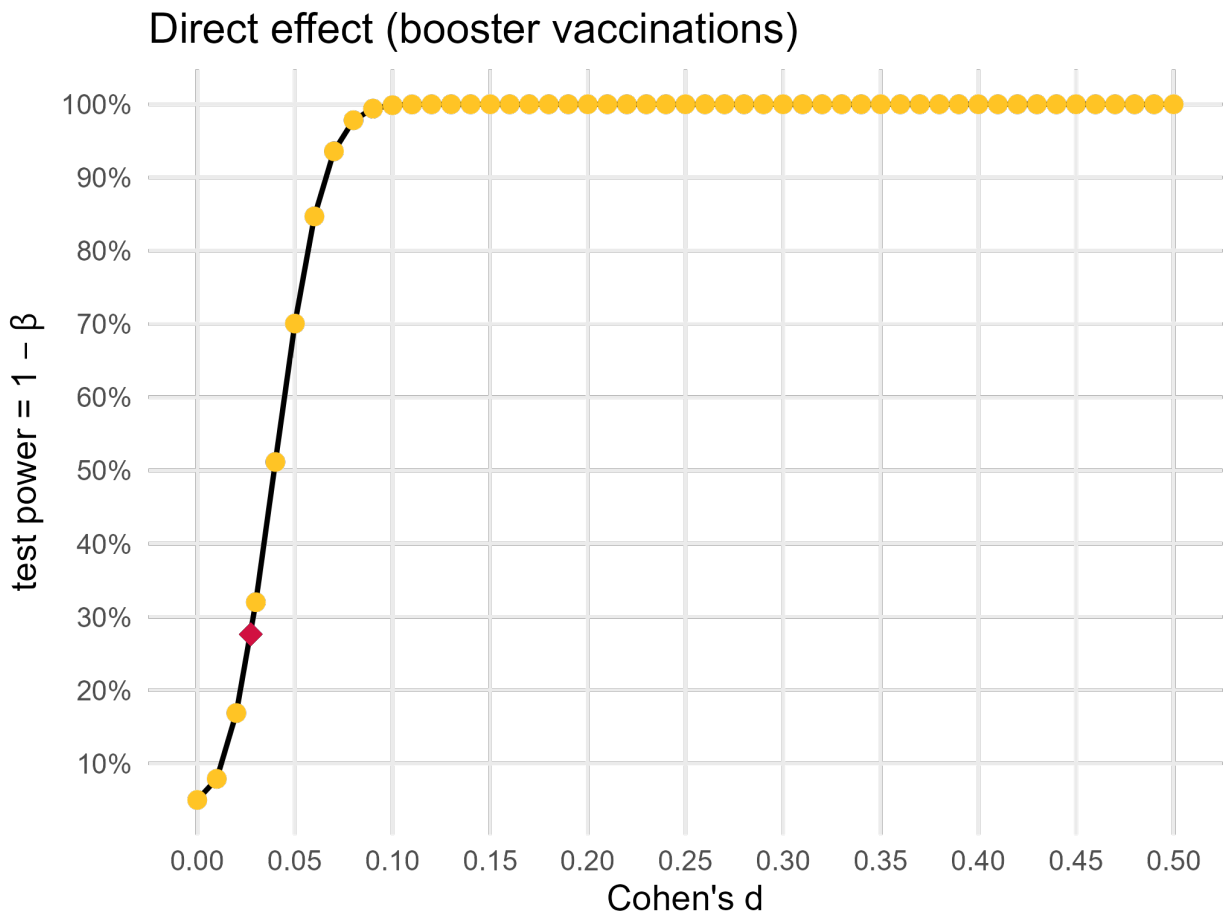


Figure S12: Sensitivity analysis of the direct effect on booster vaccinations.

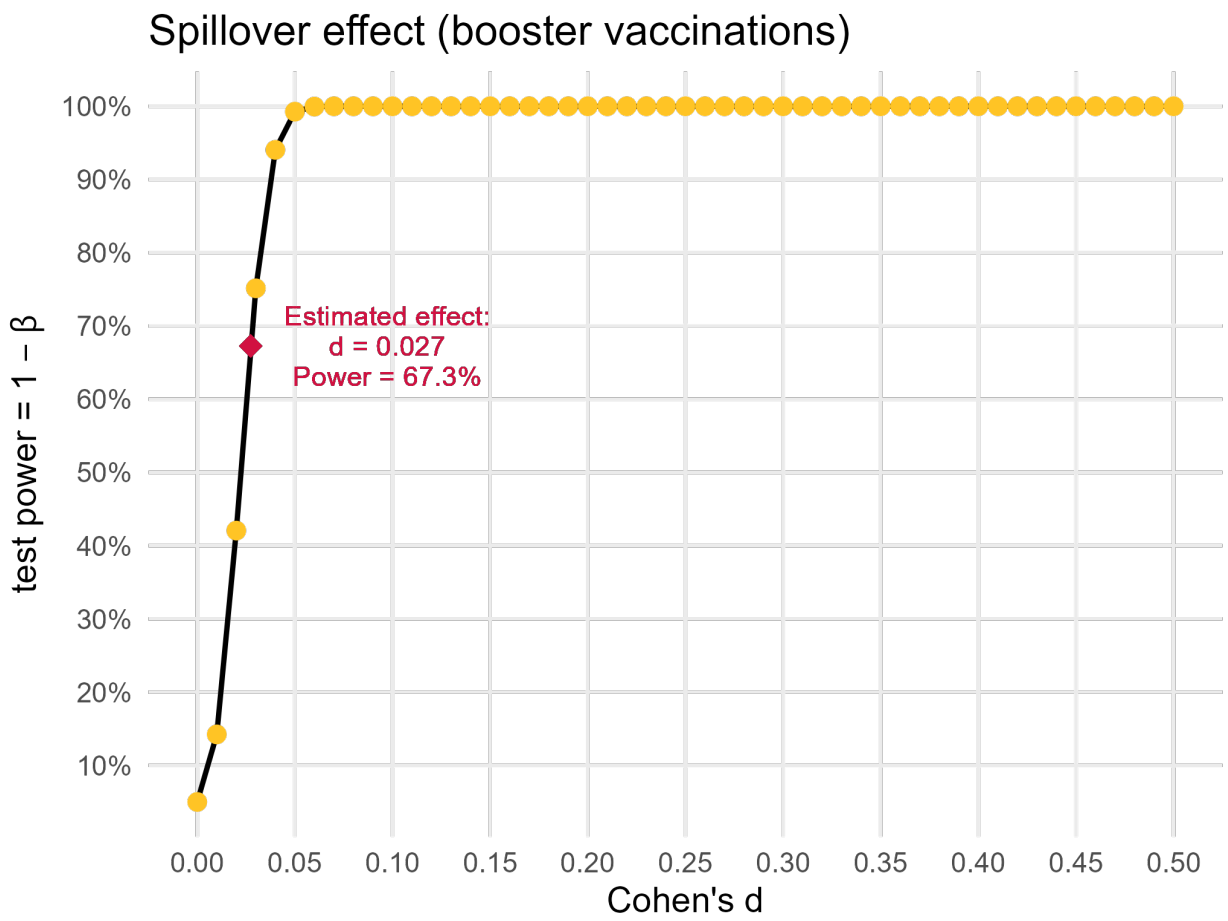


Figure S13: Sensitivity analysis of the spillover effect on booster vaccinations.

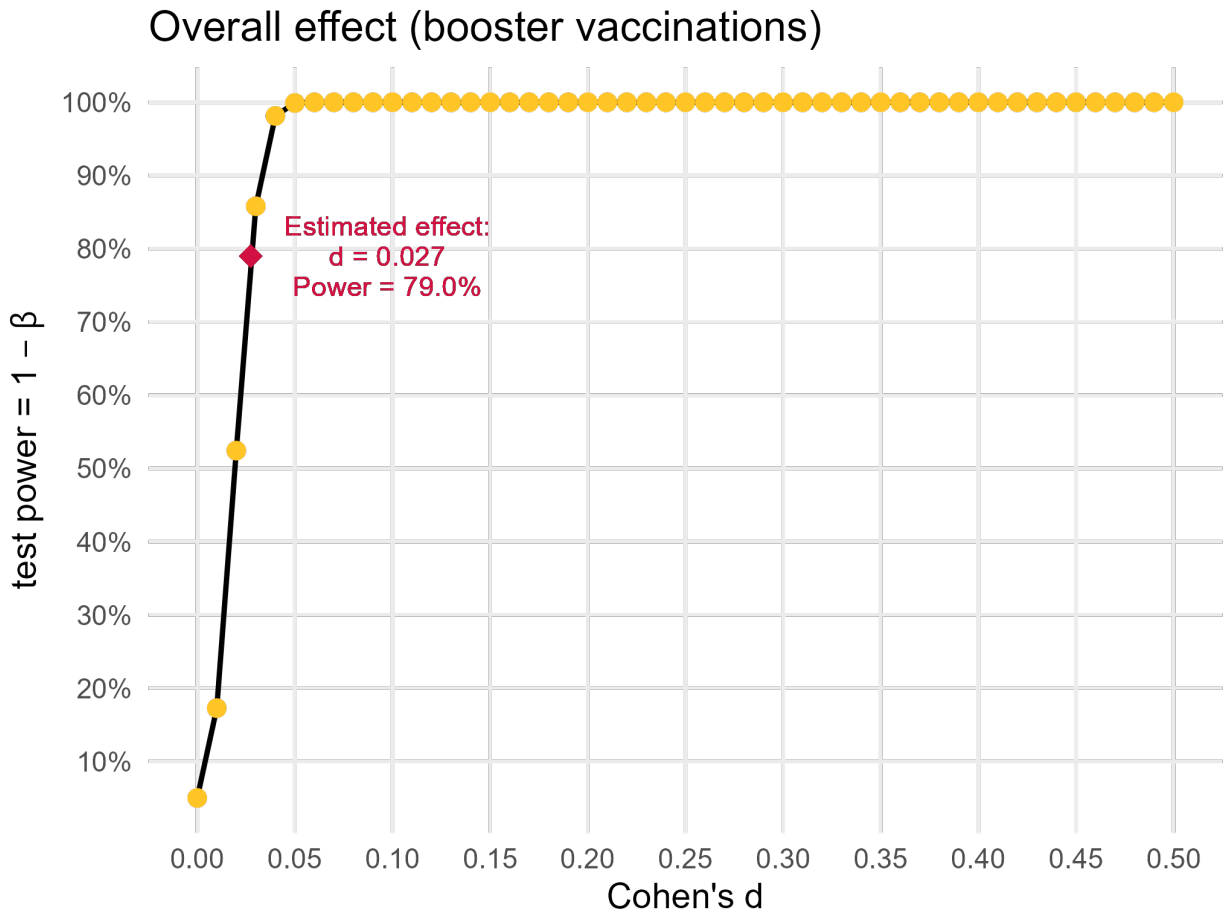


Figure S14: Sensitivity analysis of the overall effect on booster vaccinations.

To supplement this power analysis with a more precise formulation of the magnitude of effects we would be able to observe with our data, we applied randomization inference to the observed outcome data. For every treatment effect we present in Figure 4, we randomly permute the treatment assignment vector (in the same way that random assignment was initially conducted [i.e., block and cluster random assignment]) for 1,000 iterations and calculate the synthetic "null" treatment effect. This results in a synthetic null distribution of observed treatment effects, which yields a standard randomization inference p-value (as seen in Section 5.3). We then iteratively incorporate a treatment effect in the synthetic distribution by increasing the probability of the synthetic outcome variable for synthetic treatment participants by 0.1 percentage points. (Individuals who are assigned to the synthetic control keep their randomly permuted outcomes.) We then calculate the treatment estimator of interest (either the direct, overall, or spillover effect) in this synthetic data. We repeat for 1,000 iterations to generate a synthetic distribution where the true effect is known. We compare the observed "true" treatment effect to this synthetic distribution of a known synthetic effect. We repeat until the probability of observing the true treatment effect in our observed data is less than .05 in the synthetic distribution. The results for all of our estimators are presented below in Table S40.

Table S40: Probability of observing the treatment effect.

Effect Type	**If true effect size is at least**	**How unlikely is it to appear as observed in our data?**
Direct effect (information uptake)	0.010	3.5%
Direct effect (vaccinations)	0.003	4.6%
Spillover effect (vaccinations)	0.003	4.6%
Overall effect (information uptake)	0.015 (with a constant spillover effect of 0.0005)	2.9%
Overall effect (vaccinations)	0.003 (with a constant spillover effect of 0.002)	1.6%

7 Section 7: Data and Code Availability

To avoid inadvertently disclosing participants' sensitive health and socio-demographic information, we do not provide the full data publicly. However, we provide a reduced dataset that includes the outcome variables (i.e, vaccination and information uptake), treatment condition (i.e., treatment arm and cluster representative status), and a cluster ID that represents the individual's household.

Additionally, we salted the data to increase anonymity. Specifically, we randomly selected three vaccinated and three unvaccinated participants in the treatment and control arms, respectively, and inverted their vaccination uptake record. We did this twice: once for clusters with 25 or more cohabitants and once for clusters with less than 25 cohabitants. This procedure results in 24 inverted outcome measures. This way, interested researchers can reproduce our main results without exposing participants to potential de-anonymization. While the replication data excludes covariates and all our reported results include covariate adjustment, we note that covariate adjustment does not meaningfully alter any results. Furthermore, we provide the full R analysis code and its output in an additional document.

8 Section 8: Ethics

The study was approved by Zeppelin University's ethics committee on October 29th, 2022. The committee waived individuals' informed consent as it relates to 1) being included in the study and 2) the collection of data as to whether or not an individual was observed consenting to be vaccinated at one of the vaccination events. In strict compliance with the guidelines of the German public health authorities, all vaccination events included a mandatory consultation by a doctor prior to vaccination and the participant's signed confirmation of consent to get vaccinated. The Zeppelin University ethics committee waived participants' informed consent as it relates to being included in the study and the collection of data as to whether or not a participant was observed consenting to be vaccinated at the public event.

9 Section 9: Pre-registered Analysis Plan

We pre-registered the data collection, analysis, and hypotheses at the Open Science Framework (https://osf.io/2trmb/?view_only=4d4cd0da2f86424a812c6b76282c78e5) and at ISRCTN (study record #59503725). The Open Science Framework pre-registered analysis plan is reproduced here in full, for convenience.

9.1 Study Information

9.1.1 Hypotheses

H1-1 (direct or "total" effect #1): Offering a set of monetary incentives increases the number of unique website visits (on a municipal webpage with information on COVID-19 vaccination) among treated subjects in treatment clusters compared to equivalent subjects in control clusters (i.e., in each control cluster, the person who would have received the treatment letter had they been assigned to a treatment cluster).

H1-2 (direct or "total" effect #2): Offering a set of monetary incentives increases COVID-19 vaccinations (tracked at seven city vaccinations events) among treated subjects in treatment clusters compared to equivalent subjects in control clusters (i.e., in each control cluster, the person who would have received the treatment letter had they been assigned to a treatment cluster).

H2 (spillover effect): Offering a set of monetary incentives increases COVID-19 vaccinations among untreated individuals in treatment clusters compared to untreated subjects in control clusters (minus the individual who would have received a treatment letter in the control cluster if they were assigned to a treatment cluster).

H3-1 (overall effect #1): Offering a set of monetary incentives increases unique website visits among treated and untreated individuals in treatment clusters compared to untreated subjects in control clusters.

H3-2 (overall effect #2): Offering a set of monetary incentives increases COVID-19 vaccinations among treated and untreated individuals in treatment

9.2 Design Plan

9.2.1 Study Type

Experiment - A researcher randomly assigns treatments to study subjects, this includes field or lab experiments. This is also known as an intervention experiment and includes randomized controlled trials.

9.2.2 Blinding

No blinding is involved in this study.

9.3 Study Design

The design is a blocked, two-stage (individual and household) randomized controlled field experiment.

9.3.1 Randomization

The study will be conducted in a medium-sized city in Germany. Treatment assignment follows a two-stage randomization process. In the first stage, all housing addresses within the city will be randomized into two groups (i.e., intervention group and control group), blocked by household size. In the second stage, one resident of each address in both groups will be randomly selected (a "cluster representative"). The cluster representatives in the intervention group only will be sent the intervention letter. All remaining residents will be sent the control letter.

The randomization will be conducted in R (version 4.1.1) using the function "sample" with the seed set to 2021.

9.4 Sampling Plan

9.4.1 Existing Data

Registration prior to creation of data.

9.4.2 Data collection procedures

The study will be conducted in the city of Ravensburg, Germany. All residents of the city (age 18 and older) will be subject to the trial (as per official residency records). The only notable exception is that the study excludes the inmates of a local prison (n=257) and the second largest cluster of the city (n=75). Treatment assignment follows a two-stage randomization process. In the first stage, all housing addresses within the city will be randomized into two groups (i.e., intervention group and control group), blocked by household size. In the second stage, one resident of each address in both groups will be randomly selected (a "cluster representative"). The cluster representatives in the intervention group only will be sent the intervention letter. All remaining residents will be sent the control letter.

The treatment group will receive a letter from the mayor of the city inviting them to get a COVID-19 vaccination at one of seven public vaccination events. The letter offers two financial incentives. If the recipient is not already vaccinated and gets vaccinated at the vaccination events, they will receive 20 Euros (in the form of a shopping voucher). Additionally, another 20 Euro incentive is offered if more than 900 city residents get vaccinated at one of the seven vaccination events. If the recipient is already vaccinated, they can still get the second incentive if they give the treatment letter to someone else and that individual brings it to one of the seven vaccination events, gets vaccinated, and more than 900 city residents get vaccinated during one of the seven vaccination events.

The control group will get the same letter as the treatment group but without the offer of monetary incentives.

9.5 Sample Size

The sample consists of all residents of the city (n = 41,549) clustered into 10,033 addresses. The only notable exception is that the study excludes the inmates of a local prison (n=257) and the second largest cluster of the city (n = 75).

9.5.1 Sample size rationale

The sample size is determined by the population of the city. Based on simulation-based power calculations, we are able to detect a minimal direct or "total" effect of 2.6 percentage points, a spillover effect of 1.5 percentage points, and an overall effect of 1.3 percentage points with approximately 80% statistical power (alpha = 0.05). This calculation is based on the assumption that 20% of the population of the city is not eligible for a primary vaccine or booster and that eligibility is uniformly distributed across individuals. We project that up to 20% of the control group will get vaccinated at these events.

9.5.2 Stopping rule

The effect of the treatment will be assessed by measuring how many people get vaccinated at the seven public vaccination events on 2021-11-13, 2021-11-19, 2021-11-20, 2021-11-26, 2021-11-27, 2021-12-10, and 2021-12-11.

9.6 Variables

9.6.1 Manipulated variables

The treatment group will receive a letter from the mayor of the city inviting them to get a COVID-19 vaccination at one of seven public vaccination events. The letter offers two financial incentives. If the recipient is not already vaccinated and gets vaccinated at the vaccination events, they will receive 20 Euros (in the form of a shopping voucher). Additionally, another 20 Euro incentive is offered if more than 900 city residents get vaccinated at one of the seven vaccination events. If the recipient is already vaccinated, they can still get the second incentive if they give the treatment letter to someone else and that individual brings it to one of the seven vaccination events, gets vaccinated, and more than 900 city residents get vaccinated during one of the seven vaccination events.

The control group will get the same letter as the treatment group but without the offer of monetary incentives

9.6.2 Measured variables

Primary measures:

Information uptake is measured by whether a participant visited the informational website mentioned in the letter (unique website visits). The website is prominently displayed in both treatment and control letters. Each URL and QR-code is unique to the individual recipient.

Vaccination uptake is measured via an on-site record of administered vaccinations during the seven public vaccination events. The records will distinguish between primary vaccines (i.e., first and second doses) and boosters.

9.6.3 Covariate measures

These data will be collected from municipal administrative records and will be used as covariates to increase precision

- Age
- Sex
- Indicator for whether the individual is a non-German national

9.6.4 Additional measures

To determine an individual's close social network, we use address data from municipal administrative records to identify shared addresses and generate unique clusters. As an alternative specification, we use shared last name and shared address to generate a secondary cluster id. The treatment letters given to other residents by fully-vaccinated treatment letter recipients will be collected at the public vaccination events to track which subjects were referred to get vaccinated by the treatment-letter recipient.

We will also collect information (self-reports) whether patients are part of a vulnerable high-risk group which the German federal government prioritizes in the vaccine distribution.

9.7 Analysis Plan

9.7.1 Statistical models

We will use OLS regression with a treatment indicator, block fixed effects, and control covariates (consisting of a binary indicator for female, an indicator for non-German nationality, age, and age squared) in all subsequent analyses unless indicated otherwise. To ensure robustness, a secondary analysis will regress outcomes against the treatment indicator without controls.

H1-1 (direct or "total" effect #1): Information uptake regressed against treatment group for only cluster representatives, with robust standard errors.

H1-2 (direct or "total" effect #2): Vaccination uptake regressed against treatment group for only cluster representatives, with robust standard errors.

H2 (spillover effect): Vaccination uptake regressed against treatment group for all individuals who are not cluster-representatives, with robust clustered standard errors.

H3-1 (overall effect #1): Information uptake regressed against treatment group for all individuals, with robust clustered standard errors.

H3-2 (overall effect #2): Vaccination uptake regressed against treatment group for all individuals, with robust clustered standard errors.

We will use p-values (threshold = 0.05) associated with the treatment indicator coefficient in our regressions.

9.7.2 Data exclusion

Aside from the exclusion of the two largest clusters, since we use block fixed effects, the one subject in a single-person household that was randomly determined to be in a block size of one will be dropped from analysis.

If a person gets multiple shots (e.g., first and second dose), we will count them only once - namely, for their first shot.

9.7.3 Exploratory analysis

For the analysis, all residents living at the same address will be treated as members of the same social network. We will test an alternative operationalization of a social network by considering all residents with the same last name who reside in the same address cluster as members of the same social network.

We will also explore whether the hypothesized effects differ by German vs. non-German nationals.

To better map out social influence, we will also compare which letters were given to individuals living in the same address as the treatment-letter recipient, which were given to individuals in the same address with the same last name, and which were given to individuals at a different address.

Additionally, we will conduct analysis restricted to

1) first doses (excluding second doses and booster shots),

2) first and second doses (excluding booster shots), and

3) boosters (excluding first and second doses).

To explore whether the treatment effects in H1-2, H2, and H3-2 are driven by an increase in booster shots, we will perform the analyses as described in H1-2, H2, and H3-2 but using a multinomial logistic regression with no record of vaccination equal to zero, primary vaccination equal to 1, and booster vaccination equal to 2. A Wald test will be used to determine if the treatment coefficient in terms of primary vaccination is statistically significantly different from that of boosters.

There is a fair chance that in early November, the Standing Committee on Vaccination at the Robert-Koch Institute – Germany's scientific body to provide recommendations for vaccine schedules – may recommend boosters to either the whole population, or some larger subset of the population (e.g., based on age and risk factors). As of 2021-11-02, they recommend boosters only for

people aged 70 or older and people with a high risk of severe consequences of a COVID-19 infection (e.g., immunocompromised patients). The extension of these guidelines may occur in the middle of our collection of outcomes. If this occurs, for the population affected by the announcement, we will also analyze the interaction of treatment assignment with the timing of this announcement (as a binary indicator =0 when outcomes are collected before the announcement and =1 when outcomes are collected after the announcement).

In addition, we will also examine heterogenous treatment effects by single event-sessions (analyzing Friday/Saturday events as a single event-session, which makes 4 event-sessions in total). This means we will estimate the above-described models for hypotheses 1-3 by each event-session separately. We will do so cumulatively (i.e., event-session #1, event-sessions #1 + #2, ..., event-sessions #1 + #2 + #3 + #4) and non-cumulatively (i.e., net vaccination rates for each event-session), and correct for multiple testing. The event-session-analysis will depict before and after comparisons of potential public health announcements and regulations by the Standing Committee on Vaccination (STIKO), the Federal Government, and/or the State of Baden Württemberg that occur during the study period.

We will run additional exploratory analyses to include any additional vaccination events that the city communicates in conjunction with the vaccination campaign, but which were not announced in the letters. We compare them to results of the vaccination events announced in the letters.

To have a better understanding of all indirect costs of using monetary incentives to encourage vaccinations, we will also examine whether clusters assigned to treatment (as compared to clusters assigned to control) are more likely to contact the municipal government with feedback and if that feedback is more negative. (For a discussion of reactance to similar pro-social nudges, see Mann 2010). Civil servants managing the hotline about the vaccination program rank how positive the call is on a Likert scale and we also track all email inquiries. We will have the emails coded in a similar way and we will then analyze these outcomes as recommended in Coppock (2019).

Coppock, A. (2019). Avoiding post-treatment bias in audit experiments. Journal of Experimental Political Science, 6(1), 1-4.

Mann, C. B. (2010). Is there backlash to social pressure? A large-scale field experiment on voter mobilization. Political Behavior, 32(3), 387-407.

9.8 Other

This preregistration is an update to a previous one. We published it prior to collecting any data. The update was necessary to add exploratory analyses to capture regulatory interventions by government bodies that might occur during the study period and could affect the effects of the treatment as well as to capture decisions made by the involved city. We, therefore, added the last three paragraphs under "exploratory analysis", and specified that people getting vaccinated multiple times will only be counted once (see "Data exclusion").

10 Section 10: Additional Analyses

10.1 Regression Table of Impact on Information Uptake

Table S41: Treatment effects on information uptake. OLS regression estimates of the effects of the treatment on information uptake.

	Direct effect		**Spillover effect**		**Overall effect**	
	(1) Unajdusted	(2) Adjusted	(3) Unajdusted	(4) Adjusted	(5) Unajdusted	(6) Adjusted
Treatment	0.339	0.337	-0.101	-0.106	0.006	0.000
	(0.339)	(0.339)	(0.152)	(0.152)	(0.138)	(0.138)
Age		0.138*		0.142***		0.140***
		(0.056)		(0.022)		(0.021)
Age^2		-0.001**		-0.001***		-0.001***
		(0.001)		(0.000)		(0.000)
Female		-0.168		0.103		-0.002
		(0.389)		(0.169)		(0.151)
Non-German		-0.116		0.357		0.283
		(0.703)		(0.252)		(0.237)
Std. Errors	Hetero-skedasticity-robust	Hetero-skedasticity-robust	by address cluster	by address cluster	by address cluster	by address cluster
n	10,032	10,032	31,516	31,516	41,548	41,548

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

10.2 Regression Tables of Main Results

Table S42: Treatment effects. OLS regression estimates of the effects of the treatment on vaccinations.

	Direct effect		**Spillover effect**		**Overall effect**	
	(1) Unajdusted	(2) Adjusted	(3) Unajdusted	(4) Adjusted	(5) Unajdusted	(6) Adjusted
Treatment	-0.239	-0.247	-0.278	-0.269	-0.269+	-0.263+
	(0.269)	(0.269)	(0.170)	(0.169)	(0.149)	(0.148)
Age		0.033		0.014		0.017
		(0.048)		(0.023)		(0.021)
Age^2		0.000		0.000		0.000
		(0.000)		(0.000)		(0.000)
Female		0.112		-0.018		0.045
		(0.310)		(0.150)		(0.124)
Non-German		1.235*		0.961**		1.031***
		(0.587)		(0.304)		(0.278)
Std. Errors	Hetero-skedasticity-robust	Hetero-skedasticity-robust	by address cluster	by address cluster	by address cluster	by address cluster
n	10,032	10,032	31,516	31,516	41,548	41,548

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S43: Treatment effects by type of vaccination. Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations. Effects are estimated for primary vaccinations (first and second dose) and booster vaccinations separately.

	Primary vaccination			**Booster vaccination**		
	(1) Direct Effect	(2) Spillover Effect	(3) Overall Effect	(4) Direct Effect	(5) Spillover Effect	(6) Overall Effect
Treatment	0.056	0.021	0.032	-0.319	-0.295*	-0.299**
	(0.142)	(0.124)	(0.103)	(0.233)	(0.118)	(0.108)
Age	-0.028	-0.026+	-0.026+	0.061	0.041*	0.043**
	(0.026)	(0.016)	(0.014)	(0.040)	(0.017)	(0.016)
Age^2	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.159	-0.228*	-0.184*	0.279	0.208+	0.231*
	(0.171)	(0.099)	(0.079)	(0.264)	(0.115)	(0.097)
Non-German	1.483**	1.444***	1.460***	-0.224	-0.478***	-0.420**
	(0.457)	(0.269)	(0.246)	(0.380)	(0.142)	(0.132)
Std. Errors	Hetero-skedasticity-robust	by address cluster	by address cluster	Hetero-skedasticity-robust	by address cluster	by address cluster
n	9,896	31,163	41,059	9,982	31,259	41,241

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

10.3 Pre-Registered Exploratory Analyses

Table S44: The interaction of treatment and nationality. Covariate-adjusted OLS regression estimates of the effects of the treatment on vaccinations and information uptake.

	(1) Direct effect Clicks	(2) Direct effect Vaccinations	(3) Spillover effect Vaccinations	(4) Overall effect Clicks	(5) Overall effect Vaccinations
Treatment	0.112	-0.205	-0.255	0.001	-0.248+
	(0.363)	(0.285)	(0.168)	(0.152)	(0.148)
Age	0.138*	0.033	0.014	0.140***	0.017
	(0.056)	(0.048)	(0.023)	(0.021)	(0.021)
Age^2	-0.001**	0.000	0.000	-0.001***	0.000
	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Female	-0.162	0.110	-0.018	-0.002	0.045
	(0.389)	(0.310)	(0.150)	(0.151)	(0.124)
Non-German	-1.320	1.459+	1.001*	0.287	1.080*
	(0.948)	(0.876)	(0.486)	(0.334)	(0.445)
Treatment * Non-German	2.378+	-0.444	-0.081	-0.009	-0.099
	(1.310)	(1.208)	(0.620)	(0.448)	(0.565)
Std. Errors	Hetero-skedasticity-robust	Hetero-skedasticity-robust	by address cluster	by address cluster	by address cluster
n	10,032	10,032	31,516	41,548	41,548

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table S45: Effects of the treatment on first dose vaccinations. OLS regression.

	Direct effect		**Spillover effect**		**Overall effect**	
	(1) Unajdusted	(2) Adjusted	(3) Unajdusted	(4) Adjusted	(5) Unajdusted	(6) Adjusted
Treatment	0.054	0.051	-0.055	-0.041	-0.026	-0.017
	(0.124)	(0.124)	(0.124)	(0.119)	(0.101)	(0.098)
Age		-0.014		-0.001		-0.005
		(0.023)		(0.014)		(0.012)
Age^2		0.000		0.000		0.000
		(0.000)		(0.000)		(0.000)
Female		-0.237		-0.194*		-0.168*
		(0.148)		(0.088)		(0.071)
Non-German		1.014**		1.179***		1.159***
		(0.385)		(0.259)		(0.234)
Std. Errors	Hetero-skedasticity-robust	Hetero-skedasticity-robust	by address cluster	by address cluster	by address cluster	by address cluster
n	9,884	9,884	31,114	31,114	40,998	40,998

Estimates are expressed in terms of percentage points. Standard errors in parentheses. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

We stated that we would attempt to map out the spread of letters given to others (from already fully-vaccinated to unvaccinated social contacts) via network analysis, but there were too few such transfers in our sample. Only 5 letters were given to other individuals who thereafter showed up to get vaccinated and presented the letters to vaccination officials. We do not pursue multinomial logistic regression for different types of vaccinations, as we pre-registered this analysis only in case of positive overall treatment effects. We also pre-registered that we would assess whether treatment affected residents' willingness to contact the city. Only 0.77% of citizens called the hotline of which only 27 could be matched with the main data, so we were underpowered to detect differences across conditions.

11 Section 11: The Development and Prevalence of Evidence-Based Policy-Making

Before EBPM gained traction among policy-makers, the evidence-based medicine movement demonstrated the importance of going beyond doctors' expert opinions in a series of high-profile studies (Baron, 2018; Bothwell et al., 2016). From the outset, evidence-based medicine tended to value RCTs as the most preferred method for building evidence of intervention effectiveness (Baron, 2018). High profile cases included the Salk polio vaccine RCT conducted across ~130 U.S. schools, which eventually led to the eradication of polio in the country (American Academy of Pediatrics, 2022). The RCT was eventually made a legal requirement in many countries in the world as part of the standardized process for testing the efficacy and safety of new medical interventions (Bothwell et al., 2016).

Policy RCTs existed in academia decades before the term EBPM was coined,[5] but EBPM, and its prioritization of RCTs, began to coalesce into its modern form when influential governments began to codify EBPM into law and establish new EBPM institutions (McGann et al., 2018). In the US, the Education Science Reform Act of 2002 established the Institute for Educational Sciences, which worked with social scientists to measure educational program efficacy (Whitehurst, 2018). The Institute prioritized RCTs and, unsurprisingly, the number of educational RCTs grew eightfold between 2002 and 2016 (Connolly et al., 2018). Concurrently, non-government organizations, such as J-PAL, came to specialize in evaluating policies in the context of international development and rapidly grew in size and prominence; the number of impact evaluations conducted by J-PAL grew by almost an order of magnitude between 2002 and 2010 (de Souza Leão & Eyal, 2019). In 2010, the UK government created the Behavioral Insights Team (BIT) (John, 2014), which specialized in RCTs to evaluate policy effectiveness. The US followed suit shortly after; in 2015 the White House established the Social and Behavioral Sciences Team, which now resides within the US's General Services Administration and has been renamed as the Office of Evaluation Sciences (OES) (Office of Evaluation Science, 2023a). However, perhaps one of the prominent examples of the influence of EBPM in government came in the form of the Evidence Act. In 2018, a highly-polarized US Congress passed one of the few influential, overwhelmingly-bipartisan pieces of legislation in recent times: The Foundations for Evidence-based Policymaking Act of 2018. This law compelled all federal agencies to develop program evaluation capacity and to rigorously evaluate programs and policies (Office of Evaluation Science, 2023b).

Although, initially, there was considerable controversy as to what research counts as "evidence" in EBPM (Heinrich, 2007), because of RCTs prominence in evidence-based medicine, they eventually came to play a central role in EBPM as well (Baron, 2018; DellaVigna et al., 2022; Hjort et al., 2021). Many governments across the world have established public and private EBPM institutions that championed RCTs as the "gold standard" method for evaluating the causal impact of policies and programs (see, behavioral insights teams (Mukherjee & Giest, 2020) and public sector innovation labs (McGann et al., 2018).[6] And while alternative research methods have been used within the

[5] Early forms of EBPM included social policy, welfare, employment and education RCTs (Baron, 2018).

[6] A partial list of these organizations includes BIT in the UK (Haynes et al., 2012), the Fonds d'experimentation pour la jeunesse in France (Ministère de l'Éducation nationale, 2023), Ontario Behavioural Insights Unit in Canada (Government of Ontario, 2023), MineduLab in Peru (J-PAL, 2023), the Behavioural Economics Team and the New South Wales Behavioural Insights Unit in Australia, eMBeD at the World Bank (World Bank, 2023). In the US, organizations like OES and the White House's Office of Science and Technology Policy (Office of Evaluation Science, 2023c; The White House, 2014) prioritize RCTs on the federal level, but there is also a medley of state and local government equivalents such as the California Policy Lab or The Lab @ DC. In the Global South, private

context of EBPM and highlighted in prominent policy documents (The White House, 2020), the RCT has generally been the dominant research method for building evidence of policy effectiveness (Cairney & Oliver, 2017; DellaVigna et al., 2022; Evidence-Based Policymaking Collaborative, 2016; Monaghan & Ingold, 2019; Pearce & Raman, 2014).[7]

There is no shortage of EBPM process guides targeting policy-makers; many of these guides mention RCTs as part of the impact evaluation stage and their value in building evidence in favor or against a particular policy (Al-Akhali, 2020; Evidence-Based Policymaking Collaborative, 2016; The Pew Charitable Trusts & MacArthur Foundation, 2014; Tseng, 2015; Zanti & Thomas, 2021). However, the possibility of followup evaluation after scaling is infrequently mentioned and when it is, as with, for example, by the Evidence-Based Policymaking Collaborative (2016), there is little guidance as to the concrete steps a policy-maker should take to incorporate evaluation when scaling up a policy intervention.[8] And while recent works, such as (List, 2022) have focused on the possible pitfalls of scaling policy, the many different approaches to evaluating whether a policy has successfully scaled may be too technical for policy-makers. As such, since the simplicity of the RCT has helped it become a commonly-used tool in EBPM, we, similarly, propose incorporating the population-level RCT as a new step in the EBPM process before scaling up policy.

institutions such as the Busara Center for Behavioral Economics (headquartered in Kenya) (Busara, 2023) and the Center for Utilizing Behavioral Insights for Children (Save the Children International, 2023) work closely with governments to design, execute, and analyze policy RCTs (Silan et al., 2023).

[7] It's worth noting that the meta-analysis (Bothwell et al., 2016) and, more recently, the megastudy (Milkman et al., 2021) are considered by many as stronger evidence than any given RCT. However, both approaches usually require prohibitively more resources and/or much longer wait times before policy is implemented. As such, meta-analyses are not prominently featured in EBPM guides that target policymakers (Al-Akhali, 2020; Evidence-Based Policymaking Collaborative, 2016; The Pew Charitable Trusts & MacArthur Foundation, 2014; Zanti & Thomas, 2021) and are therefore outside of the scope of this paper.

[8] Contrast this with the plethora of step-by-step guides for conducting RCTs (Haynes et al., 2012).

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